

FINITE ORDINARY MINIMAL MODEL PROGRAMS ON COMPACT KÄHLER FOURFOLDS

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ABSTRACT. We prove that a globally Weil- $\mathbb{Q}$ -factorial compact Kähler klt fourfold pair with effective rational boundary and canonical rational line bundle admits a finite ordinary minimal model program starting on the given pair. It ends at a nef model when the adjoint class is pseudo-effective and at a Mori fibre space otherwise.

CONTENTS

1. Introduction	1
2. Classes, currents, and exceptional divisors	5
3. Ordinary steps on the given space	10
4. Termination for ordinary terminal fourfold pairs	16
5. Scaling and a limiting terminal pair	23
6. The limiting model and the perturbation contradiction	26
7. The two endpoints	31
Appendix A. Why analytic local factoriality is different	32
References	34

1. INTRODUCTION

The minimal model program seeks a birational model whose canonical class, or more generally whose adjoint $K_X + \Delta$, is nef. When no such model is compatible with pseudo-effectivity, its expected outcome is a Mori fibre space: a fibration along which the adjoint is negative. Each birational step contracts an extremal ray or replaces a small contraction by its flip. Constructing the individual steps and choosing a sequence that reaches an endpoint are separate problems.

For compact Kähler fourfolds, we prove that a suitable choice of ordinary negative steps gives a finite program. It starts on the given rational klt pair in the global Weil-divisor $\mathbb{Q}$ -factorial category and reaches the endpoint dictated by pseudo-effectivity. The absence of a global ample polarization makes both the contractions and the finiteness argument analytic. A second issue is categorical: a divisor or reflexive sheaf defined on the entire compact space carries information different from a divisor germ on an arbitrary analytic neighbourhood. The proof keeps this distinction, and keeps the actual canonical rational line bundle, throughout the program.

1.1. Category and the finite-program theorem. All varieties are reduced irreducible second-countable complex analytic spaces. A Kähler form on a normal space has smooth strictly plurisubharmonic local potentials in local embeddings. Projectivity of a morphism means the existence of a relatively ample holomorphic line bundle.

Definition 1.1. A normal compact space X is *globally Weil $\mathbb{Q}$ -factorial* if every prime Weil divisor defined on the whole of X is $\mathbb{Q}$ -Cartier and some positive reflexive power of its canonical sheaf ω_X

is a line bundle. It is *globally strongly $\mathbb{Q}$ -factorial* if every coherent rank-one reflexive sheaf $\mathcal{F}$ on X has an invertible positive reflexive power $\mathcal{F}^{[m]} = (\mathcal{F}^{\otimes m})^{**}$.

The Weil convention is that of [5, Definition 2.1]; the strong convention is used in [13, Section 2]. The strong condition implies the Weil condition. Neither definition imposes $\mathbb{Q}$ -factoriality on every analytic local ring. The distinction matters even for projective varieties; an example is given in Appendix A.

A rational line bundle is an element of $\text{Pic}(X) \otimes_{\mathbb{Z}} \mathbb{Q}$. An equality of rational line bundles means an isomorphism of holomorphic line bundles after a common positive integral multiple. We write K_X additively for the canonical sheaf. When a canonical Weil divisor is given, we keep that representative and transport it through the program. We will also prove the intrinsic variant in which only the canonical rational line bundle is specified. In that variant, identities of adjoints mean isomorphisms of holomorphic line bundles after a common positive multiple. Relative canonical divisors are defined using compatible local canonical generators. In particular, a numerical identity alone never specifies such a divisor.

For an effective rational boundary Δ , put $D = K_X + \Delta$. We use *log discrepancies* throughout:

$$a(E; X, \Delta) = 1 + \text{coeff}_E(K_W - p^*(K_X + \Delta))$$

on a smooth model $p : W \rightarrow X$ carrying E . A pair is klt when all these numbers are positive. It is terminal when it is klt and they are greater than one for every exceptional divisor over X . The same convention will apply to the auxiliary real and generalized pairs.

Let $H_{\text{BC}}^{1,1}(X)$ denote real Bott–Chern cohomology of closed $(1, 1)$ -forms with local potentials, and let $\overline{\text{NA}}(X)$ be the closed cone of positive currents of bidimension $(1, 1)$ modulo their pairings with these classes. A class is *nef* if it belongs to the closure of the Kähler cone, and *pseudo-effective* if it contains a closed positive $(1, 1)$ -current with local potentials. These terms for a rational line bundle refer to its first Chern class. A class is *big* if it contains a Kähler current, meaning a closed current that dominates a positive multiple of a Kähler form. For a projective contraction $f : X \rightarrow Z$, write $\rho(X/Z)$ for the dimension of the space of global real line-bundle classes modulo zero degree on every contracted curve, and put

$$\rho_{\text{BC}}(X/Z) = \dim_{\mathbb{R}}(H_{\text{BC}}^{1,1}(X)/f^*H_{\text{BC}}^{1,1}(Z)).$$

An *ordinary D -negative step* is a projective divisorial contraction, or the flip of a projective small contraction, of a D -negative extremal ray of $\overline{\text{NA}}(X)$. The contraction has connected fibres and a normal compact Kähler target; its relative rank is one and $-D$ is relatively ample. For a flip the transformed adjoint is ample over the same target. The boundary is pushed forward in a divisorial step and strictly transformed in a flip. A *Mori fibre space* has the same contraction properties and a base of strictly smaller dimension.

Theorem 1.2 (Finite ordinary programs). *Let X be a normal globally Weil $\mathbb{Q}$ -factorial compact Kähler fourfold, with a canonical Weil divisor K_X , and let $\Delta \geq 0$ be a rational divisor such that (X, Δ) is klt. Then there is a finite sequence of ordinary negative steps*

$$(X, \Delta) = (X_0, \Delta_0) \dashrightarrow \cdots \dashrightarrow (X_n, \Delta_n).$$

Every X_i is normal, compact Kähler and globally Weil $\mathbb{Q}$ -factorial; every (X_i, Δ_i) is klt, and $D_i = K_{X_i} + \Delta_i$ is $\mathbb{Q}$ -Cartier. The canonical divisors are the transforms of the specified datum. The following alternatives hold.

- (i) *If $D = K_X + \Delta$ is pseudo-effective, then D_n is nef. The composite $\phi : X \dashrightarrow X_n$ extracts no prime divisor, and*

$$a(E; X, \Delta) \leq a(E; X_n, \Delta_n)$$

for every prime divisor over the two models, with strict inequality for every prime on X contracted by ϕ .

- (ii) *If D is not pseudo-effective, there is a projective surjective morphism $g : X_n \rightarrow S$ with connected fibres, where S is normal compact Kähler, $\dim S < 4$, $\rho(X_n/S) = 1$, and $-D_n$ is g -ample.*

For each negative contraction, including g , the relative Bott–Chern dimension is one. If X is globally strongly $\mathbb{Q}$ -factorial, so is every X_i .

The same conclusions hold in the intrinsic formulation with the canonical rational line bundle in place of a chosen canonical Weil divisor. In either formulation the first model is exactly X .

Theorem 1.2 gives an affirmative solution of the finite-program form of the rational klt fourfold minimal model conjecture in this global compact Kähler category. The quantifier is existence of one finite ordinary program; its proof uses auxiliary models without replacing the first model X . In the pseudo-effective case, the endpoint assertion is nefness, with the discrepancy properties displayed above.

1.2. History and the additional argument. The projective minimal model program supplies the geometric framework: negative extremal contractions, flips and the comparison of discrepancies. Kawamata–Matsuda–Matsuki developed the four-dimensional terminal termination argument using discrepancies and the loss of algebraic surface classes [15, Theorem 5-1-15 and Lemmas 5-1-16–5-1-17]; Kollár–Mori give a systematic account of the birational tools [16]. Fujino extended the fourfold termination method to canonical pairs with rational boundary [8, 9]. His addendum identifies a necessary correction to the low-discrepancy count: all valuations over certain smooth boundary surfaces must be treated together, rather than only the divisor obtained by the first blowup.

In the compact Kähler setting, Höring–Peternell constructed minimal models for non-uniruled threefolds [14]. Das–Hacon–Păun proved the fourfold minimal model theorem for dlt pairs whose adjoint is $\mathbb{Q}$ -linearly equivalent to an effective divisor [4, Theorem 1.1]. Their work also develops analytic finite-generation and birational tools. Fujino’s relative analytic MMP supplies ordinary cone, base-point-free and flip theorems for projective morphisms between complex analytic spaces [10]. These are relative projective constructions; neither the source nor the final model is required to be projective over a point.

Generalized pairs provide a way to carry a nef class on a higher model through the program. Das–Hacon–Yáñez developed their compact Kähler threefold theory and the analytic language used here [5]. Hacon–Xie’s cone and transcendental base-point-free theorems supply contractions in the compact Kähler category [13, Theorems 1.3–1.4]. Their generalized MMP with scaling terminates when the boundary plus nef part is big [13, Theorem 1.5]; their Theorem 1.1 gives Mori fibre models in the non-pseudo-effective case and good models in its stated big cases. These results furnish the steps, the auxiliary programs, and the finiteness of marked models away from scaling parameter zero that we use below.

Matsumura–Zhong prove termination with Kähler scaling for strongly $\mathbb{Q}$ -factorial compact Kähler klt pairs whose pseudo-effective adjoint has numerical dimension zero. They also obtain minimal models after a small crepant strong $\mathbb{Q}$ -factorial modification in that case [18, Theorems 6.1 and 6.3].

The remaining issue for Theorem 1.2 is the pseudo-effective adjoint without a bigness or effectivity assumption. We follow the limiting-model strategy of Birkar [1, Theorem 1.2 and Section 4]. In that strategy, termination for a limiting terminal pair is combined with rational decomposition of a nef real adjoint and a sufficiently small perturbation. Birkar already works with real boundaries in the projective category. Here the required replacements concern compact Kähler models, potentially transcendental scaling data, and varying projective contraction bases.

Three points make this transfer possible. First, the trace of the one initial Kähler form is identified as a positive current with local potentials on every relevant model. Second, we prove ordinary terminal fourfold termination with effective real boundary by combining the corrected low-discrepancy argument with compact analytic cycle classes. Third, actual line bundles numerically trivial over the relevant birational contractions descend without an additional tensor power. The

latter fact preserves the Cartier indices of finitely many rational nef adjoints and therefore the positive intersection gap used in the perturbation argument. None of these auxiliary constructions changes the initial model of the ordinary program.

1.3. Proof strategy and organization. Sections 2–4 establish the common analytic, birational and termination tools. The proof of Theorem 1.2 then has four stages.

1. Fix a Kähler form on the original space and run ordinary steps with its transformed generalized nef data as a scaling direction. If there were infinitely many steps, marked-model finiteness would force their scaling parameters to tend to zero.
2. Extract the stabilized finite set of low-discrepancy places. On a fixed auxiliary model the boundaries decrease to an effective ordinary terminal boundary.
3. Run a finite program for that limiting terminal pair and express its nef adjoint as a positive combination of rational nef adjoints. Their fixed Cartier indices give a uniform positive lower bound for nonzero degrees on curves.
4. A sufficiently small generalized perturbation has a finite program on which all these rational adjoints are crepant. At its endpoint, descent to an earlier flip base makes the originally negative adjoint trivial on its contracted ray, a contradiction.

Section 7 identifies the endpoint and concludes the finite-program proof. All extractions occur on auxiliary models; the ordinary sequence itself begins on X .

Section 2 constructs the trace and comparison tools, and Section 3 supplies ordinary steps and exact bundle descent. Section 4 proves the terminal theorem before it is used. Sections 5 and 6 carry out the four stages above, and Section 7 identifies the endpoints. Appendix A gives a smooth projective example showing why analytic local $\mathbb{Q}$ -factoriality cannot replace the global convention in the theorem. Figure 1 records the dependencies of this argument.

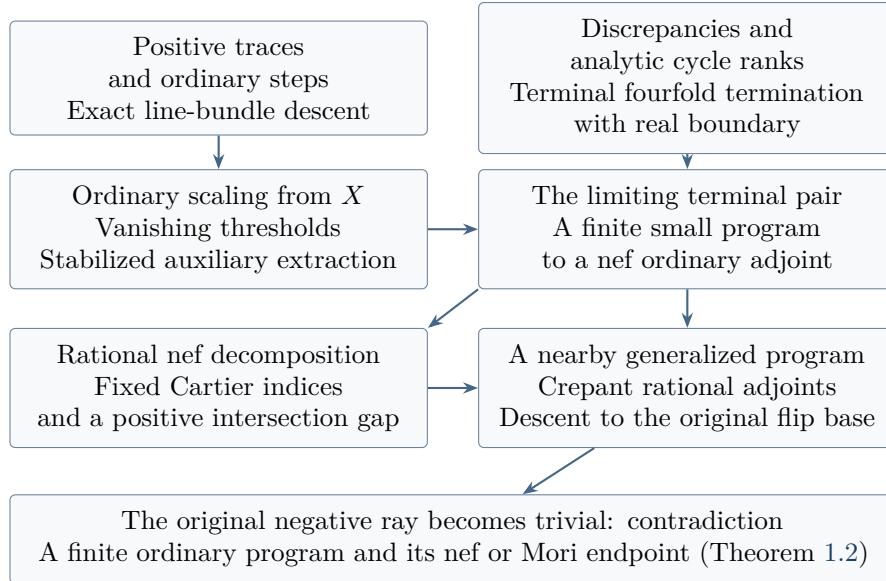


FIGURE 1. The finite-program argument. The ordinary sequence starts on X . Extraction and the limiting and nearby programs are auxiliary comparisons. Exact bundle descent preserves the indices used in the positive intersection gap; the last descent occurs on a common resolution, as in Figure 2.

2. CLASSES, CURRENTS, AND EXCEPTIONAL DIVISORS

The scaling argument transports one Kähler form from the original space through several birational models. Its trace need not be smooth, or even have local potentials without further argument. We establish the precise trace relation, its positivity, and the comparison rules used below. Equalities between Bott–Chern classes will be distinguished from equalities between divisors or currents.

2.1. Pullback of classes. We use the following cohomological descent theorem. A space is in *Fujiki’s class $\mathcal{C}$* if it is bimeromorphic to a compact Kähler manifold.

Lemma 2.1 (Pullback and descent). *Let $f : T \rightarrow Z$ be a proper surjective morphism with connected fibres between normal compact analytic spaces with rational singularities. Suppose that T is in class $\mathcal{C}$ and that either the general fibre is rationally connected or $R^j f_* \mathcal{O}_T = 0$ for every $j > 0$. Then*

$$f^* : H_{\text{BC}}^{1,1}(Z) \longrightarrow H_{\text{BC}}^{1,1}(T)$$

is injective, and its image consists exactly of the classes having degree zero on every f -vertical curve. If T and Z are Kähler, a class on Z is nef if and only if its pullback is nef.

Proof. The pullback assertion is [13, Lemma 2.39]; nef descent is [4, Lemma 2.38]. In particular, the first assertion applies to any proper bimeromorphic morphism between compact spaces with rational singularities whose source is in class $\mathcal{C}$: its general fibre is a point. For a projective klt log Fano contraction, relative vanishing supplies the higher-direct-image hypothesis once rationality of the base has been established. $\square$

2.2. Positive currents on a normal model. An f -exceptional real divisor is a finite real linear combination of prime divisors whose images have codimension at least two. We next give the descent statement needed when an exceptional correction has either sign. Pseudo-effectivity requires a positive current with local plurisubharmonic potentials; we do not assume that an arbitrary positive current on a singular space has them.

Lemma 2.2 (Exceptional descent). *Let $p : U \rightarrow T$ be a resolution of a normal compact Kähler space, with U compact Kähler. Let $\alpha \in H_{\text{BC}}^{1,1}(T)$ and let E be a p -exceptional real divisor. If $p^* \alpha + [E]$ is pseudo-effective, then α is pseudo-effective. No sign condition on E is needed. Pseudo-effective classes pull back to resolutions, and nef classes on T are pseudo-effective.*

Proof. Choose a smooth representative θ of α with local smooth potentials. On the smooth compact Kähler space U , a positive representative of the given class has the form

$$S = p^* \theta + [E] + \text{dd}^c v \tag{2.1}$$

for a global distribution v . Remove from T its singular locus and the image of the exceptional locus, obtaining a smooth open set T° whose complement has codimension at least two. The map p is an isomorphism there. Equation (2.1) gives a global θ -plurisubharmonic function v° on T° . We do not assert that v is quasi-plurisubharmonic along E .

On a coordinate neighbourhood where $\theta = \text{dd}^c \rho$, extend $\rho + v^\circ$ first across the removed codimension-two subset of the regular locus, and then across the singular locus. The first extension is the plurisubharmonic Hartogs theorem; the second is its normal-space version of Grauert–Remmert, recalled in [3, Section 2.1, pp. 927–928]. In the latter statement, every plurisubharmonic function on the regular locus of a normal space extends uniquely to the whole space; local upper boundedness at the singular points is not a further hypothesis to be deduced from v . We apply it only after obtaining a plurisubharmonic function on the entire regular locus. These extensions are unique. On overlaps their differences are the original smooth pluriharmonic differences of the functions ρ . Subtracting ρ therefore gives a global potential v_T , and

$$\theta + \text{dd}^c v_T \geq 0$$

is a current with local plurisubharmonic potentials representing exactly α . This proves descent, including for signed E .

For pullback, compose local plurisubharmonic potentials with the resolution. They cannot become identically $-\infty$ on a nonempty open subset, since a resolution is an isomorphism on a dense open set. Thus they define the pulled-back positive current; see [3, Section 2.1, p. 928].

Finally, if α is nef, then $p^*\alpha$ is nef on U . For a Kähler form ω_U , choose positive representatives of $p^*\alpha + \varepsilon[\omega_U]$. Their masses are bounded as $\varepsilon \downarrow 0$, so a weak limit is a positive current in $p^*\alpha$. On the smooth Kähler manifold U it has local plurisubharmonic potentials. The first part, with $E = 0$, descends this current to the required class on T . $\square$

2.3. Generalized pairs and negativity. We recall the generalized-pair language in the form needed here [5, Section 2.1 and Definition 2.7]. A closed b -(1,1) *current* is a compatible collection of closed (1,1)-currents on proper birational models: pushforward along a morphism between models recovers the current on the lower model. It *descends* to a model U if its trace there and all higher traces have local potentials, and the classes of the higher traces are pullbacks of its class on U . It is *b-nef* if that class is nef on some such model. We also use the relative version for real combinations of line-bundle data on a carrier projective over a specified base. Here nefness means nonnegative degree on curves contracted to that base. The structure equation and discrepancy definition below are unchanged.

A generalized pair on T consists of a boundary $B \geq 0$, a b-nef datum $\mathbf{M}$, and a log resolution $\nu : U \rightarrow T$ on which the datum descends, together with an actual real divisor B_U such that $\nu_*B_U = B$ and

$$[K_U + B_U] + [\mathbf{M}_U] = \nu^*L, \quad L \in H_{\text{BC}}^{1,1}(T). \quad (2.2)$$

The support of B_U may be taken to have simple normal crossings. Negativity makes this structure boundary unique; passing to higher resolutions gives the other structure boundaries. The generalized log discrepancy of a prime P on such a resolution is

$$a(P, T, B + \mathbf{M}) = 1 - \text{coeff}_P B_U.$$

The pair is generalized klt, or *gklt*, if all these discrepancies are positive, and generalized log canonical, or *glc*, if they are all nonnegative. The *gklt locus* is the open subset where all discrepancies of primes centred there are positive; its complement is the generalized non-klt locus. We use the word *terminal* for an ordinary or generalized klt pair whose discrepancies at all exceptional primes are strictly greater than one. Negative coefficients are permitted in B_U , although the boundary B on the model is effective. Generalized klt spaces have rational singularities [5, Theorem 2.19].

The negativity lemma concerns actual divisors, not arbitrary Bott–Chern classes. We record also the strict form needed for fourfold termination.

Lemma 2.3 (Negativity and strict comparison).

- (i) Let $f : T \rightarrow Z$ be a proper bimeromorphic morphism of normal complex spaces and let E be a real Cartier divisor such that $-E$ is f -nef. Then $E \geq 0$ if and only if $f_*E \geq 0$. Consequently an f -exceptional, f -numerically trivial real Cartier divisor is zero.
- (ii) If, in addition, f has connected projective fibres and $E \geq 0$ is f -anti-nef, then every fibre is either contained in $\text{Supp } E$ or disjoint from $\text{Supp } E$.
- (iii) Consider a nontrivial small birational diagram

$$T \xrightarrow{f} Z \xleftarrow{f^+} T^+$$

of normal compact spaces, with projective contractions on both sides. Let $B \geq 0$ be a real boundary, let B^+ be its strict transform, and suppose the ordinary adjoints $D = K_T + B$ and $D^+ = K_{T^+} + B^+$ are real Cartier, with $-D$ relatively ample on the left and D^+ relatively

ample on the right. The two exceptional loci have the same image $A \subset Z$. On a common resolution $p : V \rightarrow T$, $q : V \rightarrow T^+$,

$$p^*D - q^*D^+ = F, \quad F \geq 0, \quad (2.3)$$

where F is an actual divisor exceptional over both models. For every prime divisor P over these spaces,

$$a(P, T^+, B^+) \geq a(P, T, B),$$

and the inequality is strict if the centre of P on either side is contained in that side's exceptional locus.

- (iv) For the diagram in (iii), if $\dim T = 4$, then $\dim A \leq 1$ and

$$\dim \text{Exc}(f) + \dim \text{Exc}(f^+) \geq 3.$$

- (v) The discrepancy conclusions in (iii) also hold for generalized pairs with the same fixed b -nef datum and transformed boundaries in a projective small diagram, provided the source log class has negative degree on every curve contracted on that side, and the target log class has positive degree on every curve contracted on the other side. They hold likewise for a projective divisorial contraction $f : T \rightarrow T'$ when the target boundary is the pushforward, the b -datum is unchanged, and the source log class has negative degree on every f -vertical curve. In this divisorial case strictness holds at every prime whose source centre lies in $\text{Exc}(f)$. This part also applies when the fixed b -datum is nef only over a specified base.

Proof. Part (i) is the analytic negativity lemma [13, Lemma 2.5]; apply it to both signs for the last assertion. For (ii), points of a connected projective fibre can be joined by a chain of irreducible curves. If the fibre met both $\text{Supp } E$ and its complement, some curve in such a chain would meet the support without being contained in it. Its intersection with the effective real Cartier divisor would be strictly positive, contrary to f -anti-nefness. The assertion for a real Cartier divisor follows locally from positive combinations of effective rational Cartier divisors, as explained below.

For (iii), let G be the normalization of the graph in $T \times_Z T^+$, with projections p_G, q_G and map $h : G \rightarrow Z$. Compatible canonical data give the actual real Cartier divisor

$$F_G = p_G^*D - q_G^*D^+.$$

This difference is defined locally by compatible meromorphic canonical generators and therefore glues, even when the canonical sheaves were specified as $\mathbb{Q}$ -line bundles. Smallness makes it exceptional over both sides. It is effective by (i), applied over T : on a p_G -vertical curve the degree of $-F_G$ is the degree of $q_G^*D^+$, which is nonnegative.

For every h -vertical curve C one has

$$F_G \cdot C < 0. \quad (2.4)$$

Indeed, normalization is finite onto the graph, so the two projections cannot both contract C . The nonzero degrees contributed by p_G^*D and $-q_G^*D^+$ are both negative. An effective real Cartier divisor has nonnegative degree on a curve outside its support. For completeness, this fact and its strict pullback analogue hold even if its individual prime components are not $\mathbb{Q}$ -Cartier: locally express it in the real span of finitely many Cartier divisors. Vanishing of coefficients outside its support and nonnegativity of the remaining coefficients are rational linear conditions. The resulting rational polyhedral cone expresses it as a positive real combination of effective $\mathbb{Q}$ -Cartier divisors. The usual effective Cartier assertions apply to each term; summing the nonnegative local intersection numbers gives the asserted global degree inequality. In particular, the pullback has positive coefficient at every prime whose centre is contained in the support.

The morphism p_G has connected fibres because T is normal; hence $h = fp_G$ has connected fibres. Its fibres are projective because G is finite over the graph in the fibre product. Each point of a positive-dimensional fibre lies on a curve in that fibre. Equation (2.4) therefore places the entire fibre

in $\text{Supp } F_G$. To identify the relevant fibres, suppose f is an isomorphism over an open subset of Z . There the log divisor on T^+ is pulled back from Z , by smallness and agreement in codimension one. Relative ampleness excludes an f^+ -vertical curve there; projectivity, connectedness and normality then make f^+ an isomorphism there as well. Interchanging the sides proves that their exceptional images coincide. Thus $h^{-1}(A) \subset \text{Supp } F_G$.

Pull F_G back to a resolution on which P appears. Its coefficient at P is exactly $a(P, T^+, B^+) - a(P, T, B)$, with our log-discrepancy convention. Effectivity gives monotonicity, and the support assertion gives the claimed strictness. This proves (iii).

For (iv), a small exceptional locus in a fourfold has dimension at most two and has positive-dimensional fibres, so its image has dimension at most one. Moreover $F_G \neq 0$ by (2.4). A component of its support has dimension three and maps finitely into $\text{Exc}(f) \times_Z \text{Exc}(f^+)$. The dimension of this product is at most the sum of the dimensions of its two factors. This proves (iv).

For (v), choose a common smooth resolution $p : V \rightarrow T$, $q : V \rightarrow T'$ carrying the fixed b-datum and projective over the common contraction base. Such a resolution is obtained by taking projective modifications of the graph that resolve the maps to a carrier of the datum. Subtract the actual structure boundaries to obtain a real Cartier divisor $F = B_{T,V} - B_{T',V}$. Its coefficients are the differences of generalized log discrepancies, and cancellation of the fixed b-part gives

$$[F] = p^*L_T - q^*L_{T'}.$$

Let $h : V \rightarrow Z$ be the morphism to the common contraction base; in the divisorial case $Z = T'$ and $q = h$. It is projective by this choice and has connected fibres because it is bimeromorphic onto the normal space Z . The boundary-pushforward condition makes F exceptional over Z . The relative signs of the log classes give $F \cdot C \leq 0$ on every h -vertical curve, with strict inequality whenever $p(C)$ or, in the small case, $q(C)$ is a curve. Thus (i) gives $F \geq 0$.

If y lies in an exceptional image, a vertical curve downstairs can be lifted to a curve on V dominating it: use a component of its projective inverse image and cut by relatively ample hyperplanes. Such a curve has negative F -degree and hence lies in $\text{Supp } F$. Part (ii) now puts the entire fibre $h^{-1}(y)$ in the support. Pullback to a higher resolution has positive coefficient at every prime centred in this support. This proves all the strict discrepancy inequalities, including for a centre on the flipped side. No trace current on the normalized graph is assumed to be Cartier; the actual divisor is constructed from the structure boundaries on the smooth resolution. $\square$

2.4. A fixed nef b-part and its traces. For the finite-program proof, fix a Kähler form h on the original space X , and set $H = [h]$. It defines a positive b-current $\mathbf{H}$: for a marked birational model T , choose a common resolution $p : V \rightarrow T$, $q : V \rightarrow X$ and put

$$\mathbf{H}_T = p_*q^*h. \quad (2.5)$$

Compatibility on higher resolutions makes this independent of the choice of resolution [5, Claim 2.5]. Throughout the proof this is one fixed current datum; we do not replace it by arbitrary cohomologous currents. The class of the trace, when it exists, will be denoted by H_T .

Lemma 2.4 (The exceptional trace relation). *Let T be a normal compact Kähler space with rational singularities, marked birationally to X . Suppose that on a common smooth Kähler resolution $p : V \rightarrow T$, $q : V \rightarrow X$ there are a class $H_T \in H_{\text{BC}}^{1,1}(T)$ and an actual p -exceptional real divisor J_T satisfying*

$$p^*H_T = q^*H + [J_T]. \quad (2.6)$$

Then $J_T \geq 0$. The class H_T and the divisor J_T are unique, and the correction on a higher resolution is the pullback of J_T .

For an effective boundary B_T with real Cartier ordinary adjoint, the generalized structure boundary for the nef datum $t\mathbf{H}$ is

$$B_{T,V}(t) = p^*(K_T + B_T) - K_V + tJ_T, \quad t \geq 0. \quad (2.7)$$

In particular, decreasing t increases every generalized log discrepancy. If the generalized pair is gklt, its ordinary pair is klt; if all its exceptional log discrepancies exceed one, the ordinary pair is terminal.

Conversely, suppose $t > 0$ and a generalized pair with boundary B_T and this fixed datum $t\mathbf{H}$ is given by (2.2), with log class L and real Cartier ordinary adjoint. Then (2.6) holds with

$$H_T = \frac{L - [K_T + B_T]}{t}.$$

Proof. On every p -vertical curve, $-J_T$ has the same degree as the nef class q^*H . Lemma 2.3(i) gives $J_T \geq 0$. Subtract two possible relations. The difference of the correction divisors is p -exceptional and numerically trivial over T , hence zero. Pullback injectivity from Lemma 2.1 then gives uniqueness of the class as well.

Pulling back (2.6) to a higher resolution gives the claimed correction there by uniqueness. Conversely, if the relation is initially known on a higher resolution $r : V' \rightarrow V$, push its correction down to V . The difference between the upstairs correction and the pullback of this pushforward is r -exceptional and numerically trivial over V , so is zero. Injectivity of r^* gives the relation on V .

Equation (2.7) follows by adding K_V and the fixed nef class tq^*H to its two sides. Its coefficients give

$$a(P, T, B_T + t\mathbf{H}) = a(P, T, B_T) - t \text{coeff}_P J_T$$

on every sufficiently high resolution. This proves the discrepancy assertions.

For the converse, let B_V be the given actual generalized structure boundary and put $B_V^{\text{ord}} = p^*(K_T + B_T) - K_V$. Both push forward to B_T , so

$$J_T = \frac{B_V - B_V^{\text{ord}}}{t}$$

is an actual p -exceptional divisor. Subtract the ordinary log class from (2.2) to obtain (2.6). This constructs the divisor before applying negativity; it does not infer a divisor from an arbitrary class difference. $\square$

The converse is particularly useful for an extraction carrying a pulled-back generalized log class. It shows that the divided difference defining its trace is the same class whenever the marked model is the same, even if its boundary was obtained at a different scaling parameter.

Corollary 2.5 (Positivity of the trace). *Under the hypotheses of Lemma 2.4, the actual trace $\mathbf{H}_T$ has local plurisubharmonic potentials and represents H_T . The class H_T is big.*

Proof. The positive current $q^*h + [J_T]$ represents p^*H_T . The proof of Lemma 2.2 constructs a positive current S_T in H_T with local potentials. On the big open where p is an isomorphism and the correction is absent, it agrees with p_*q^*h . Their difference is a closed current of order zero supported in codimension at least two. Such a current vanishes: after a local embedding it has bidimension $(\dim T - 1, \dim T - 1)$, whereas its support has smaller dimension [6, Chapter III, Corollary 2.11]. Thus S_T is precisely the actual trace in (2.5). This step is needed because a trace of a positive b-current need not have local potentials a priori [5, Remark 2.6(ii)].

The class q^*H is nef and has positive top self-intersection: q^*h is semipositive and is positive on the dense open where q is an isomorphism. By [7, Theorem 0.5], it contains a Kähler current $R \geq \delta\omega_V$ for some $\delta > 0$ and some Kähler form ω_V . Fix a Kähler form ω_T and choose $C > 0$ with $p^*\omega_T \leq C\omega_V$. For $0 < \varepsilon < \delta/C$,

$$R + [J_T] - \varepsilon p^*\omega_T \geq 0.$$

Its class is $p^*(H_T - \varepsilon[\omega_T])$. Lemma 2.2 shows that $H_T - \varepsilon[\omega_T]$ is pseudo-effective. Hence H_T contains a Kähler current, as required. $\square$

2.5. Comparison with the same nef datum. The preceding results supply the positivity needed for scaling. We finish with the comparison that later excludes divisorial steps in the limiting program and identifies the two nef pullbacks in the final contradiction.

Lemma 2.6 (Small-model comparison). *Let $(T, B_T + \mathbf{M})$ and $(T', B_{T'} + \mathbf{M})$ be generalized pairs with the same b -nef datum, whose log classes are L_T and $L_{T'}$. Suppose the marked birational map $T \dashrightarrow T'$ extracts no divisors and $B_{T'}$ is the pushforward of B_T . On a common sufficiently high resolution $p : V \rightarrow T$, $q : V \rightarrow T'$, put*

$$F = B_{T,V} - B_{T',V}.$$

Then F is an actual divisor exceptional over T' , and

$$p^*L_T - q^*L_{T'} = [F], \quad \text{coeff}_P F = a(P, T', B_{T'} + \mathbf{M}) - a(P, T, B_T + \mathbf{M}). \quad (2.8)$$

If the map is small, F is exceptional over both models. In this case nefness of $L_{T'}$ implies $F \geq 0$, and nefness of both log classes implies $F = 0$. These conclusions also hold relatively over a common base. Even without smallness, equality of the pullback classes in (2.8) implies equality of the structure boundaries.

Proof. Subtract the two structure equations on V . The trace of the fixed b -datum cancels, giving (2.8). At a prime dominating a divisor on T' , the corresponding boundary coefficient on T is unchanged, so its coefficient in F is zero. This proves exceptionality over T' . Smallness gives the same assertion over T .

If $L_{T'}$ is nef, then $-F$ is p -nef, because its degree on a p -vertical curve is the degree of $q^*L_{T'}$. Negativity gives $F \geq 0$. If L_T is also nef, then F is q -nef, so applying negativity with the opposite sign gives $F \leq 0$. The same proof uses only relative nefness when the models lie over a common base. For equal pullback classes, the already constructed exceptional divisor is relatively numerically trivial, and is zero by Lemma 2.3(i). $\square$

All equalities of pulled-back log classes in this lemma are cohomological. Equality of actual currents was used only in the separate trace-identification argument. These distinctions allow the ordinary steps and the generalized auxiliary constructions to share one fixed nef datum.

3. ORDINARY STEPS ON THE GIVEN SPACE

We construct the contraction and flip of every negative analytic ray in arbitrary dimension. The input is an ordinary rational klt pair in the *global Weil-divisor* category. Generalized pairs enter only through the cone and base-point-free theorems; the steps themselves are ordinary. The main point is to preserve actual line bundles, including their fixed Cartier indices, rather than only numerical classes.

3.1. Integral descent. Bundles are written additively in numerical expressions. For example, $L - (K_T + C)$ denotes the rational line bundle obtained from L and the rational adjoint. All degrees in the following lemma are degrees on compact curves contracted by the indicated morphism.

Lemma 3.1 (Integral descent). *Let $f : T \rightarrow Z$ be a projective bimeromorphic contraction of normal complex spaces, and let L be a line bundle numerically trivial over Z . Suppose that every point of Z has a neighbourhood U on which there is an ordinary rational klt pair $(f^{-1}U, C_U)$ with rational adjoint and*

$$L - (K_{f^{-1}U} + C_U) \quad f\text{-nef over } U.$$

*Then f_*L is a line bundle and the evaluation map is an isomorphism*

$$f^*f_*L \simeq L. \quad (3.1)$$

In particular the conclusion applies if the local ordinary adjoints are relatively antiample. It also applies to a global ordinary rational klt pair (T, C) whenever $L - (K_T + C)$ is f -nef. Clearing one denominator therefore descends a rational line bundle as an actual rational line bundle.

Proof. Fix $z \in Z$ and shrink to a connected Stein neighbourhood U on which the stated pair is defined. A line bundle M on $f^{-1}U$ has a Cartier-divisor representative there: the coherent direct image f_*M has rank one over the nonempty open where f is an isomorphism, so Cartan's Theorem A supplies a nonzero section. Its pullback is a section of M that is not identically zero, and its zero divisor is Cartier. This argument takes place over U , not on the whole compact space.

Every rational line bundle N on $f^{-1}U$ is also relatively big. Indeed, choose $q > 0$ with qN a line bundle and an f -ample line bundle A . The same direct-image argument gives a nonzero section of $qN - A$, hence

$$N \sim_{\mathbb{Q}} q^{-1}A + q^{-1}E \quad (E \geq 0).$$

This is relative bigness in the analytic sense [10, Definition 2.46]. Apply it to $N = L - (K_T + C_U)$. The klt base-point-free theorem gives, after shrinking around z , relative generation of L^k for every sufficiently large integer k [10, Theorem 6.2 and Remark 6.3].

Choose finitely many such generating sections near the compact fibre $F = f^{-1}(z)$. The morphism they define on the projective reduced fibre F_{red} is constant on each irreducible component: a positive-dimensional image would yield a curve with positive degree against the pullback of the hyperplane bundle, contrary to the numerical triviality of L^k . Such a curve is obtained by successive general hyperplane sections of a component. Connectedness of F makes the images of all its components the same point. A linear combination of the generating sections is consequently nonzero at every point of F_{red} .

This section is a generator of L^k in the local ring at every point of F : its residue is nonzero, and hence its representing function is a unit. This also treats a nonreduced fibre. Properness allows us to shrink U away from the image of the section's zero set. Thus L^k is trivial on $f^{-1}U$. Apply the argument to consecutive integers k and $k + 1$, and use one common neighbourhood. Their quotient trivializes L itself. Since $f_*\mathcal{O}_T = \mathcal{O}_Z$, on this neighbourhood $f_*L \simeq \mathcal{O}_U$, and evaluation is an isomorphism. These local conclusions give (3.1) globally. No additional tensor power enters the integral conclusion. $\square$

The nef variant is important even when the adjoint has degree zero over f . For example, if $P = K_T + C$ is a rational klt adjoint, ℓP is Cartier, and P is numerically trivial over f , apply Lemma 3.1 to $L = \ell P$. Its difference from the adjoint is $(\ell - 1)P$, so the *same* index ℓ descends.

3.2. Supporting classes and projectivity. A class is *modified big* if it is the birational pushforward of a big class [13, Definition 2.8]. In particular, a big class is modified big.

We use the following consequences of the cone and transcendental base-point-free theorems [13, Theorems 1.3 and 1.4]. For a compact Kähler gklt pair with a descending nef part β , the cone $\overline{\text{NA}}(T)$ is the sum of its adjoint-nonnegative part and rays of rational curves. Only finitely many rays are needed when the boundary plus β is big. If its adjoint class α is nef and the boundary plus β is modified big, there is a Moishezon contraction $f : T \rightarrow Z$ to a normal compact Kähler space with $\alpha = f^*\gamma$ for a Kähler class γ on Z . Neither assertion requires strong factoriality. Projectivity in our weaker category will be proved from the ordinary rational adjoint.

Lemma 3.2 (Supports with a Kähler margin). *Let (T, B) be an ordinary rational klt pair on a normal compact Kähler space. Assume that B is $\mathbb{Q}$ -Cartier and that $D = K_T + B$ is a rational line bundle. If D is not nef, $\overline{\text{NA}}(T)$ has a D -negative extremal ray. Every such ray R is generated by a rational curve and has a nonempty relatively open set $\mathcal{U}_R \subset R^\perp$ of classes satisfying*

$$\alpha \text{ is nef}, \quad \overline{\text{NA}}(T) \cap \alpha^\perp = R, \quad \alpha - c_1(D) \text{ is Kähler.} \quad (3.2)$$

Proof. Fix a Kähler class ω and write $d = c_1(D)$. Cone duality applies because klt singularities are rational. The slice

$$S = \{v \in \overline{\text{NA}}(T) : \omega \cdot v = 1\}$$

is compact. Indeed, fix smooth representatives of a basis of $H_{\text{BC}}^{1,1}(T)$. Each is bounded above and below by a multiple of a Kähler form, so all its pairings on S are bounded; the slice is closed. If D is not nef, the minimum of d on S is negative. An extreme point of its minimum face gives a negative extremal ray.

Fix any such ray and let $r \in S$ be its normalized point. Choose $\varepsilon > 0$ with $(d + \varepsilon\omega) \cdot r < 0$. The descending nef part $\varepsilon\omega$ changes no discrepancy, and $B + \varepsilon\omega$ is big since B is effective and $\mathbb{Q}$ -Cartier. The finite form of the cone theorem gives

$$\overline{\text{NA}}(T) = \overline{\text{NA}}(T)_{d+\varepsilon\omega \geq 0} + \sum_{j=1}^N \mathbb{R}_{\geq 0}[C_j], \quad (3.3)$$

where the C_j are rational curves. Normalize their classes in S and discard repetitions. By extremality, r is one of these points, say r_1 . Let K be the convex hull of the others and of $S \cap \{d + \varepsilon\omega \geq 0\}$. This is compact and excludes r ; otherwise extremality of r would fail.

If K is nonempty, strict separation gives a linear functional h with $h(r) = 0$ and $h > 0$ on K , after subtracting a multiple of ω . Regard it as a Bott–Chern class by duality. For all sufficiently small $\delta > 0$, $h - \delta d$ is strictly positive on K and at r . Since $S = \text{conv}(\{r\} \cup K)$, it is strictly positive on S , hence Kähler. Thus $\alpha = h/\delta$ has the required properties. Its positive minimum on K persists under small perturbations in $R^\perp$; openness of the Kähler cone also preserves $\alpha - d$. These perturbations give $\mathcal{U}_R$. If K is empty, then $S = \{r\}$ and $-d$ is Kähler; a small neighbourhood of 0 in $R^\perp$ has the required properties. This includes the case $R^\perp = \{0\}$. $\square$

Lemma 3.3 (Relative positivity). *Let $f : T \rightarrow Z$ be a proper morphism of compact complex spaces, with T and Z Kähler. Suppose that L is a rational line bundle and*

$$c_1(L) = \kappa - f^*\gamma,$$

where κ is Kähler on T and γ is a smooth real Bott–Chern class on Z . Then L is f -ample and f is projective. Conversely, a normal space projective over a compact Kähler space is compact Kähler if it has a global relatively ample line bundle.

Proof. Clear the denominator of L and choose any smooth Hermitian metric on that line bundle. The Bott–Chern identity says that the difference between its curvature and the indicated difference of forms is dd^c of a global smooth function. Adjusting the metric by the exponential of that function gives the required curvature, with the same denominator restored at the end. Smooth functions and metrics here have their normal-space meaning in local embeddings. On every fibre its curvature is positive. Positivity of a line bundle on a compact complex space implies ampleness, and for a proper analytic map ampleness on every fibre is equivalent to relative ampleness [11, Corollary 1.12 and Definition 3.1–Remark 3.2]. This criterion does not presuppose projectivity and applies to nonreduced fibres as well. Existence of the relatively ample line bundle makes f projective.

For the converse, local relative embeddings give metrics on a relatively ample bundle with positive curvature in fibre directions. Glue their weights by a partition of unity on the base. The resulting metric retains that positivity. Add a sufficiently large multiple of a base Kähler form to its curvature. The sum is strictly positive; compactness allows one multiple to work everywhere. $\square$

3.3. The ordinary contraction and flip. We can now perform each step and preserve the category. The next proposition records both the numerical ranks of the negative contraction and the actual canonical algebra of a flip. This description identifies the transformed adjoint as an actual rational line bundle.

Proposition 3.4 (Ordinary negative steps). *Let (T, B) be an ordinary rational klt pair on a normal compact Kähler space, globally $\mathbb{Q}$ -factorial in the Weil-divisor sense, with canonical sheaf a rational line bundle. Put $D = K_T + B$. Every D -negative extremal ray $R \subset \overline{\text{NA}}(T)$ has a projective contraction $f : T \rightarrow Z$ with connected fibres and normal compact Kähler target, such that*

$$f^* H_{\text{BC}}^{1,1}(Z) = R^\perp, \quad \rho(T/Z) = 1, \quad -D \text{ is } f\text{-ample.} \quad (3.4)$$

Here ρ is the rank of global line-bundle degrees on contracted curves. The target has rational singularities and $R^j f_* \mathcal{O}_T = 0$ for $j > 0$. If $\dim Z < \dim T$, this is a Mori fibre space. Otherwise f is bimeromorphic, and precisely one of the following occurs.

- (i) The exceptional locus is one prime divisor E . Put $T' = Z$, $B' = f_* B$ and $D' = K_{T'} + B'$. Then

$$D = f^* D' + eE \quad (e > 0). \quad (3.5)$$

- (ii) The contraction is small. For an adjoint Cartier index $r > 0$, the intrinsic algebra

$$\mathcal{R}_r = \bigoplus_{m \geq 0} f_* \mathcal{O}_T(mrD) \quad (3.6)$$

is locally finitely generated, and $T' = \text{Proj}_Z \mathcal{R}_r$ is the ordinary flip. It has a small projective morphism $f^+ : T' \rightarrow Z$ with connected fibres. Put $B' = \phi_* B$ and $D' = K_{T'} + B'$, where $\phi : T \dashrightarrow T'$ is the induced small map. For a sufficiently divisible choice of r there is an actual isomorphism

$$\mathcal{O}_{\text{Proj}_Z \mathcal{R}_r}(1) \simeq \mathcal{O}_{T'}(rD'). \quad (3.7)$$

In particular D' is a rational line bundle and is f^+ -ample. Passing to a sufficiently divisible Veronese leaves T' unchanged.

In both birational cases, T' is compact Kähler and globally Weil-divisor $\mathbb{Q}$ -factorial, its canonical sheaf is a rational line bundle, and (T', B') is klt. Strong global $\mathbb{Q}$ -factoriality is preserved when imposed on T . A supplied global canonical Weil divisor is transported to a canonical Weil divisor on T' . No prime divisor is extracted. All log discrepancies weakly increase; the increase is strict for every original prime contracted by the step, and, for a flip, for every place centred in either exceptional locus.

Proof. Choose $\alpha \in \mathcal{U}_R$ from Lemma 3.2 and put $\kappa = \alpha - c_1(D)$. Insert the descending nef part κ into the ordinary pair. It changes no discrepancy, while $B + \kappa$ is big. Transcendental base-point-freeness gives $f : T \rightarrow Z$ and γ Kähler with $\alpha = f^* \gamma$. Thus $c_1(-D) = \kappa - f^* \gamma$, and Lemma 3.3 proves projectivity and relative ampleness. This establishes projectivity without strong factoriality. For every curve $C \subset T$,

$$f(C) \text{ is a point} \iff \alpha \cdot C = 0 \iff [C] \in R. \quad (3.8)$$

The rational curve generating R is contracted, so f is nontrivial.

Relative vanishing gives $R^j f_* \mathcal{O}_T = 0$ for $j > 0$ [10, Theorem 5.2]; the log Fano base has rational singularities [4, Lemma 8.8]. Lemma 2.1 now identifies $f^* H_{\text{BC}}^{1,1}(Z)$ with the classes of degree zero on all vertical curves, which is exactly $R^\perp$. The global line-bundle rank is also one, since all those curves are proportional and the rational line bundle D detects their ray. This proves (3.4). The lower-dimensional case has all the stated Mori fibre-space properties. In equal dimension, connectedness gives generic degree one, so f is bimeromorphic.

Divisorial contractions. Suppose E is an exceptional prime. It is $\mathbb{Q}$ -Cartier. If $E \cdot R \geq 0$, it would be f -nef, contradicting negativity for a nonzero effective exceptional divisor. Thus $E \cdot R < 0$. Every vertical curve lies in E , since an effective $\mathbb{Q}$ -Cartier divisor has nonnegative degree on a curve outside its support. Every positive-dimensional projective fibre is covered by curves. A zero-dimensional local component of a connected fibre would be isolated, and the birational form of Zariski's main theorem would make f an isomorphism there. Consequently $\text{Exc}(f) = E$.

For a prime Weil divisor A_Z on Z , let A_T be its strict transform. Choose $t \in \mathbb{Q}$ so $(A_T + tE) \cdot R = 0$. The coefficient is rational by taking the ratio of the two rational degrees on one integral curve. A Cartier multiple of $A_T + tE$ descends by Lemma 3.1: its difference from D is relatively ample. The descended line bundle agrees with the corresponding reflexive power of $\mathcal{O}_Z(A_Z)$ off $f(E)$, a subset of codimension at least two. Reflexivity identifies them everywhere. Hence every global prime on Z is $\mathbb{Q}$ -Cartier.

If a global canonical Weil divisor is given, push it forward. It agrees with the canonical sheaf on the big open where f is an isomorphism, so reflexivity shows that it remains a canonical Weil representative. Alternatively, start with an invertible reflexive power of ω_T , adjust it by a rational multiple of E to degree zero, and descend after clearing denominators. On the same big open the result is a reflexive power of ω_Z ; hence that power is invertible everywhere. This proves the canonical assertion without assuming a global meromorphic canonical section.

Now D' is a rational line bundle and its canonical identification away from E gives (3.5). Intersection with R gives $e > 0$. If b_E is the coefficient of E in B , then

$$a(E; Z, B') = 1 - b_E + e > 1 - b_E = a(E; T, B) > 0.$$

Pullback of the effective divisor eE gives weak increase for all other log discrepancies. Thus the target pair is klt.

The small canonical model. Suppose f is small. Finite generation of the ordinary relative log canonical ring [10, Theorem 1.18] gives local finite generation of (3.6). When canonical data are written as sheaves, this application can be made over each Stein open in Z : nonzero sections of rank-one proper direct images provide the meromorphic representatives there. Changing representatives induces compatible graded isomorphisms of the intrinsic algebra.

The ordinary analytic flip theorem [10, Theorem 1.14] constructs precisely its relative canonical model. It has no local factoriality hypothesis. Its proof glues the unique local log canonical models, giving a normal small projective $f^+ : T' \rightarrow Z$ globally. The fibres are connected because Z is normal. A finite cover of the compact base permits one common sufficiently divisible Veronese generated in degree one. Replace r by that multiple. The resulting tautological bundle is relatively ample and agrees with $\mathcal{O}_T(rD)$ on the common big open. Its reflexive extension is $\mathcal{O}_{T'}(rD')$, proving (3.7) as an actual bundle isomorphism. This also proves that D' is a rational line bundle. The relative Proj is invariant under this Veronese operation. By Lemma 3.3, T' is compact Kähler.

For a global prime A' on T' , transform it to A on T and choose $t \in \mathbb{Q}$ so $(A + tD) \cdot R = 0$. Descend a Cartier multiple by Lemma 3.1. On T' its pullback agrees on the common big open with the same multiple of $A' + tD'$. Reflexivity extends the identification; since D' is already a rational line bundle, A' is $\mathbb{Q}$ -Cartier. This proves global Weil-divisor factoriality. A given canonical Weil divisor is transported on that common open and then reflexively. In the sheaf formulation, repeat the descent argument with an invertible power of ω_T adjusted by tD ; undoing the twist by tD' proves the rational-line-bundle property of $\omega_{T'}$.

On a common smooth resolution $p : V \rightarrow T$, $q : V \rightarrow T'$, Lemma 2.3 gives the actual discrepancy divisor

$$p^*D = q^*D' + F, \quad F \geq 0, \quad (3.9)$$

exceptional over both sides, with the asserted strictness over the flipping and flipped loci. This also proves klt preservation.

Optional strong factoriality. In the divisorial case, take any coherent rank-one reflexive sheaf $\mathcal{F}$ on Z . Its reflexive pullback to T is coherent of rank one. Strong factoriality supplies an invertible reflexive power. Adjust it by a rational multiple of E , descend as above, and compare off $f(E)$. A reflexive power of $\mathcal{F}$ is therefore a line bundle.

In the small case start with such a sheaf $\mathcal{F}'$ on T' . On a common smooth resolution put

$$\mathcal{M} = (q^*\mathcal{F}'/\text{torsion})^{**}, \quad \mathcal{F} = (p_*\mathcal{M})^{**}.$$

Here $\mathcal{M}$ is a line bundle and $\mathcal{F}$ is coherent reflexive of rank one; both give the required transform on the common big open. An invertible power of $\mathcal{F}$, adjusted by a rational multiple of D , descends. Pull back the descended bundle and undo the twist by D' . After clearing denominators, reflexivity identifies the resulting line bundle with a reflexive power of $\mathcal{F}'$. Thus the strong condition is preserved. Only sheaves defined on the whole compact spaces have been used. $\square$

3.4. Transport of Bott–Chern classes. For later scaling we need to transport one fixed nef datum. The following construction uses the rank of the negative contraction only. It makes no assertion that all Bott–Chern classes on a flipped space come from the preceding space.

Lemma 3.5 (Class transport). *For a birational step of Proposition 3.4, let $f' : T' \rightarrow Z$ mean f^+ in the small case and id_Z in the divisorial case. Every $\theta \in H_{\text{BC}}^{1,1}(T)$ has a unique expression $\theta = f^*\eta + s c_1(D)$, with $\eta \in H_{\text{BC}}^{1,1}(Z)$ and $s \in \mathbb{R}$. Define*

$$\theta' = (f')^*\eta + s c_1(D'). \quad (3.10)$$

On a common resolution, $p^\theta - q^*\theta'$ is the class of an actual divisor exceptional over T' . This rule sends the class of a global Weil divisor to that of its transform or pushforward. Moreover, it propagates the fixed-nef-part trace relation of Lemma 2.4, with an actual effective exceptional correction on the new model. The same conclusions hold for ordinary real-boundary flips on strongly globally $\mathbb{Q}$ -factorial compact Kähler spaces constructed by [13, Proposition 4.2].*

Proof. Since $D \cdot R \neq 0$, subtracting a unique multiple of $c_1(D)$ makes θ vanish on R . Equation (3.4) and pullback injectivity give the unique η . On a common resolution over Z the two pullbacks of η agree. Therefore

$$p^*\theta - q^*\theta' = s[p^*D - q^*D'].$$

The expression in brackets is the actual exceptional discrepancy divisor already constructed in the proof of the proposition.

If $\theta = c_1(A)$ for a global Weil divisor A , the difference $p^*A - q^*A'$ is likewise exceptional over T' , where A' is its transform or pushforward. Subtracting the two identities represents $q^*(\theta' - c_1(A'))$ by a q -exceptional divisor. That divisor is numerically trivial over q , so negativity in both signs makes it zero. Pullback injectivity gives $\theta' = c_1(A')$.

Finally suppose the old trace H_T of a fixed nef class on an earlier model has, on a common high resolution, the relation

$$p^*H_T = p_0^*H + [J_T].$$

If $H_T = f^*\eta + s c_1(D)$, the new correction is the actual divisor

$$J_{T'} = J_T - s(p^*D - q^*D'), \quad q^*H_{T'} = p_0^*H + [J_{T'}].$$

Every old exceptional divisor remains exceptional over T' : neither kind of step extracts divisors. The resulting correction $J_{T'}$ is therefore an actual q -exceptional divisor. Its negative is q -nef because p_0^*H is nef. Negativity gives $J_{T'} \geq 0$, and uniqueness and compatibility are those of Lemma 2.4.

For the stated real-boundary extension, the cited contraction theorem supplies the negative-side Bott–Chern rank and the ordinary real flip. The difference $p^*D - q^*D'$ is still an actual exceptional real discrepancy divisor, by Lemma 2.3. The decomposition, transport formula, and correction argument above therefore apply unchanged. Rationality of D was not used in these arguments. $\square$

Remark 3.6. Proposition 3.4 starts on the given pair and can be reapplied after every finite prefix of ordinary steps. It allows every negative ray. Its construction is independent of any termination claim, and makes no assertion about the positive-side quotient $H_{\text{BC}}^{1,1}(T')/(f^+)^*H_{\text{BC}}^{1,1}(Z)$ or about equality of the full Bott–Chern ranks across a flip.

4. TERMINATION FOR ORDINARY TERMINAL FOURFOLD PAIRS

We prove termination of ordinary flips for compact Kähler fourfold pairs with effective real boundary and log discrepancy $a(E) > 1$ for every exceptional place. Two finite counts replace projectivity of the ambient fourfold: analytic cycle classes control contractions, and low-discrepancy places control flipped surfaces. The bases of successive projective contractions may vary. We first establish these counts, including their generalized-pair form needed for extraction later in the proof.

4.1. Loss of analytic cycle classes. For a compact complex analytic space V and an integer $k \geq 0$, put

$$\begin{aligned} \mathcal{C}_k(V) &:= \text{span}_{\mathbb{R}}\{[S] \in H_{2k}(V, \mathbb{R}) : S \subset V \text{ irreducible compact analytic, } \dim S = k\}, \\ c_k(V) &:= \dim_{\mathbb{R}} \mathcal{C}_k(V). \end{aligned}$$

Fundamental classes here use the complex orientation on the regular loci. Compact analytic spaces admit finite triangulations compatible with any specified finite collection of closed analytic subsets, with simplicial dimension at most twice the complex dimension. Thus $c_k(V)$ is finite. We justify this topological input after Lemma 4.1. For such a triangulation, Borel–Moore localization is the relative simplicial-chain sequence for a closed analytic subset and its complement.

If V is Kähler and S is such a subspace, then

$$\langle [\omega]^k, [S] \rangle = \int_{S_{\text{reg}}} \omega^k > 0 \quad (4.1)$$

for a Kähler form ω on V . Thus $[S] \neq 0$, including when V or S is singular.

Lemma 4.1 (Cycle loss). *Let $X \dashrightarrow Y$ be a bimeromorphic transformation of compact Kähler spaces, represented by a proper bimeromorphic diagram. Suppose it identifies*

$$U = X \setminus A \simeq Y \setminus B,$$

where A and B are closed analytic subspaces and $\dim B < k$. Then there is a surjection

$$\mathcal{C}_k(X) \longrightarrow \mathcal{C}_k(Y).$$

If A contains an irreducible compact analytic k -dimensional subspace, then $c_k(X) > c_k(Y)$.

Consequently, in a sequence of bimeromorphic transformations of n -dimensional compact Kähler spaces which extract no prime divisors, only finitely many transformations contract a prime divisor. A small transformation preserves c_{n-1} .

Proof. The Borel–Moore localization sequence for $B \subset Y$ contains

$$H_{2k}(B, \mathbb{R}) \longrightarrow H_{2k}(Y, \mathbb{R}) \xrightarrow{j_Y^*} H_{2k}^{\text{BM}}(U, \mathbb{R}) \longrightarrow H_{2k-1}(B, \mathbb{R}).$$

The outside groups vanish: the real dimension of B is at most $2k - 2$. Thus j_Y^* is an isomorphism. Compose restriction from X with its inverse to obtain a linear map

$$H_{2k}(X, \mathbb{R}) \xrightarrow{j_X^*} H_{2k}^{\text{BM}}(U, \mathbb{R}) \xrightarrow{(j_Y^*)^{-1}} H_{2k}(Y, \mathbb{R}).$$

For an irreducible k -cycle $S \subset X$ not contained in A , its restriction is the fundamental class of $S \cap U$. Closing its image in Y gives its strict transform: analyticity follows by taking the proper image of the corresponding strict transform in the given diagram. The displayed map therefore sends $[S]$ to this transformed cycle class. It sends $[S]$ to zero when $S \subset A$.

Conversely, no k -dimensional subspace of Y is contained in B . Its strict transform in X consequently maps to its fundamental class. This proves the asserted surjection of cycle spans. If $S \subset A$ has dimension k , its nonzero class, by (4.1), is in the kernel. This proves the strict inequality.

For a transformation extracting no prime divisor, choose the common isomorphic open so that the complement in the target has codimension at least two. Every contracted source prime lies in the source complement. Apply the result with $k = n - 1$ at every step. The nonnegative integer c_{n-1}

cannot decrease infinitely often. For a small transformation both complements have codimension at least two, so applying the result in both directions gives equality. $\square$

This argument does not assert surjectivity on full homology, and does not compare the Bott–Chern spaces of the two models.

Topological justification. We give the abstract-space reduction for the finite compatible triangulation used above. Teissier’s compatible Whitney stratification [22, Chapter III, Propositions 1.5 and 2.2.2] has finitely many strata on a compact space. Choose finitely many analytic charts $z_j : U_j \hookrightarrow \mathbb{C}^{N_j}$ and nonnegative smooth cutoffs ρ_j compactly supported in these charts whose positivity sets cover the space. Extend ρ_j and $\rho_j z_j$ by zero. The map

$$x \mapsto (\rho_j(x), \rho_j(x)z_j(x))_j$$

is a topological embedding into a Euclidean space. Near a point where $\rho_j > 0$, its components extend smoothly to the ambient chart and the j -th block has the smooth left inverse $(t, w) \mapsto w/t$. It is therefore locally an ambient smooth embedding, which preserves the Whitney conditions. Mather’s control data [17, Section 7, Proposition 7.1, and Section 8] and Goresky’s triangulation theorem [12, Section 5] now give a compatible triangulation, smooth on each stratum. Compactness makes it finite, and smoothness on strata gives the dimension bound. Borel–Moore localization is then the relative simplicial-chain sequence for a closed analytic subset and its complement. $\square$

4.2. Low places on compact log smooth data. We use generalized discrepancies in the sense of Section 2. A place is called exceptional over X when its center on X has codimension at least two. The generalized klt statement below supplies the finite extraction set in Section 5. Its coordinate estimate and finite centre-blowup argument also enter the terminal surface analysis in this section.

Lemma 4.2 (Low places and their projective realization). *Let $(X, B + \mathbf{M})$ be a generalized pair on a compact normal complex space, with effective real boundary of finite support and nef $b(1, 1)$ data descending to a model projective over X . Fix a projective log resolution $p : W \rightarrow X$ carrying those data.*

- (i) *If the pair is generalized klt, there is $\epsilon > 0$ such that every prime divisor over X has log discrepancy at least ϵ , and only finitely many exceptional places have*

$$a(E; X, B + \mathbf{M}) < 1 + \epsilon.$$

- (ii) *If the pair is generalized terminal and $b = \max(\{0\} \cup \{\text{coeff}_D B\})$, there are only finitely many exceptional places with*

$$a(E; X, B + \mathbf{M}) < 2 - b.$$

In each assertion the finite set of exceptional places can be represented simultaneously by prime divisors on a smooth model projective over X . In particular, any prescribed finite set of exceptional places with $a(E; X, B + \mathbf{M}) \leq 1$ for a generalized klt pair has such a realization. This conclusion does not use an extraction or termination theorem.

Proof. Let $L \in H_{\text{BC}}^{1,1}(X)$ be the generalized log class and let $M_W = [\mathbf{M}_W]$ be the nef trace class on W . Write the crepant formula as

$$[K_W + \Delta] + M_W = p^*L. \tag{4.2}$$

This is the crepant boundary identity defining generalized discrepancies; it is an equality of Bott–Chern classes when the nef data are transcendental. The boundary Δ is an actual real divisor with simple normal crossing support. Since the nef data descend to W , every further generalized discrepancy is the ordinary discrepancy for (W, Δ) . The coefficients of Δ need not be nonnegative. We may enlarge its listed simple normal crossing divisor by components of coefficient zero. Here a closed stratum means an irreducible component of an intersection of listed components; it is smooth, and is the closure of its open stratum.

We give the local estimate and the realization argument explicitly; they are the log smooth calculations underlying [16, Proposition 2.36] and [2, Proposition 2.8]. Write $\Delta = \sum_j d_j D_j$ and put $w_j = 1 - d_j$. Suppose first that every w_j is positive, and set

$$\delta = \min(\{1\} \cup \{w_j\}_j) > 0.$$

Let E be exceptional over a smooth model with this log smooth boundary. At a general point of its center C , let $c = \text{codim } C$, and choose local parameters $x_1, \dots, x_c$ for C so that the boundary components containing C are $x_1 = 0, \dots, x_r = 0$. For $v = \text{ord}_E$, the Jacobian calculation gives

$$a(E; W, \Delta) \geq \sum_{j=1}^r (1 - d_j) v(x_j) + \sum_{j=r+1}^c v(x_j). \quad (4.3)$$

Indeed, on a smooth model containing E , with local parameter t for E , write $x_j = t^{v(x_j)} u_j$ at its general point. In a top exterior differential at most one factor can use the term involving dt that lowers the order by one. Thus the relative Jacobian has order at least $\sum_{j=1}^c v(x_j) - 1$. Adding one and subtracting the boundary orders proves (4.3). The remaining coordinates along C are holomorphic and contribute no negative orders.

In particular every discrepancy is at least δ : for a divisor already on the smooth model it is its listed weight, or one if it is not listed; for an exceptional divisor use (4.3). If C is not a stratum of the listed simple normal crossing divisor, then $c > r$. When $r > 0$, the right hand side is at least $\delta + 1$; when $r = 0$, it is at least $c \geq 2 \geq \delta + 1$. Hence

$$C \text{ not a stratum} \implies a(E; W, \Delta) \geq 1 + \delta. \quad (4.4)$$

Now fix a place E with discrepancy strictly below $1 + \delta$. If it is not already a divisor on W , its center is a stratum by (4.4). Blow up that closed stratum. The center is smooth, the blowup is projective, and the new listed boundary is again simple normal crossing. If the center is the intersection of the components indexed by J , then the new exceptional component has weight

$$w_{\text{new}} = \sum_{j \in J} w_j. \quad (4.5)$$

Thus all weights remain at least δ , so the same argument applies on the next model. It excludes a non-stratum center at every stage until E becomes a divisor.

This process is finite. While E is exceptional, a blowup of its smooth center of codimension $c \geq 2$ changes its ordinary empty-boundary discrepancy by

$$a(E; W_{\text{new}}, 0) = a(E; W_{\text{old}}, 0) - (c - 1)v(\mathcal{I}_C).$$

The subtracted quantity is a positive integer, whereas an ordinary log discrepancy over a smooth space is a positive integer. Consequently only finitely many such blowups are possible. Thus E is obtained by successive stratum blowups, which is the meaning of a *toroidal place* here. This also realizes it on a projective model without first assuming that the original model on which E was given was projective over X .

For completeness, there are only finitely many such toroidal places. A toroidal prime over a fixed irreducible stratum is determined by a primitive nonzero vector $(m_j)_{j \in J}$ of nonnegative integers in the normal boundary directions. Its discrepancy is

$$\sum_{j \in J} m_j w_j. \quad (4.6)$$

These statements follow also directly from the successive stratum blowups: a blowup inserts the sum of the relevant coordinate rays, and the discrepancy transforms by (4.5). The vector determines the monomial order and is unchanged when a boundary equation is multiplied by a unit. There are finitely many irreducible strata on the compact model. Since each $w_j \geq \delta$, an upper bound T on

(4.6) bounds every m_j by T/δ . This leaves finitely many vectors and hence finitely many places. Add the finitely many prime divisors already on W which are exceptional over X .

For a generalized klt pair, all weights in (4.2) are positive. The preceding argument with $\epsilon = \delta$ proves (i) and its projective realization assertion. In particular $a \leq 1$ lies strictly below $1 + \delta$.

No rationality of the weights was used: their fixed positive minimum bounds the nonnegative integer vectors. Thus the conclusion applies to real boundaries and real nef data.

For a generalized terminal pair, the coefficient of every p -exceptional component in Δ is negative, since its log discrepancy is greater than one. The other coefficients are the original boundary coefficients and are at most $b < 1$. All the weights on W are therefore at least $1 - b$. Repeat the argument with $\delta = 1 - b$ and $T = 2 - b$ to prove (ii). Subsequent stratum blowups preserve this lower bound by (4.5).

Finally, to represent the entire finite set at once, take the dominating component of the fiber product of the finitely many projective realizations over W , normalize, and resolve projectively. The strict transforms of the designated prime divisors remain divisors on this common model. Its composite map to X is projective. $\square$

Remark 4.3. The strict inequalities in Lemma 4.2 are necessary. For example, in $\mathbf{P}^n$, $n \geq 3$, let H be a hyperplane with coefficient $b \in [0, 1)$. Blowing up any codimension-two linear subspace contained in H gives log discrepancy $2 - b$ for $(\mathbf{P}^n, bH)$. There are infinitely many such places, although this pair is terminal. With empty boundary their discrepancy is two. Below we count places strictly below a threshold and use a step at which a place reaches that threshold to obtain a strict decrease.

4.3. Surface centres of low-discrepancy places. The terminal termination argument must allow many places of discrepancy below two above a moving boundary surface. After recalling smoothness in codimension two, Lemma 4.5 identifies the infinite families that will be harmless. Excluding only the first blowup above each surface would be incorrect; the correction in [9, Proposition 3.1 and Lemma 3.2] is essential.

Lemma 4.4. *Let (T, Φ) be an ordinary klt pair on a normal complex fourfold, with effective real boundary, K_T a $\mathbb{Q}$ -line bundle, and $a(E; T, \Phi) > 1$ for every exceptional prime divisor E over T . Then T is smooth in codimension two.*

Proof. Dropping the effective boundary shows that T has terminal singularities. Suppose its singular locus has a codimension-two component. Near a general smooth point of this component choose a holomorphic projection to $\mathbb{C}^2$ submersive along it. Take a general fibre transverse both to the regular locus and, on a fixed log resolution, to the exceptional divisor and its strata. These transversality conditions hold after shrinking and choosing a general value. The resulting surface is a complete intersection in the Cohen–Macaulay space T , since klt singularities are rational, and is regular outside isolated points. It is therefore normal.

Adjunction of a local Cartier pluricanonical power, first on the regular locus and then by reflexivity, restricts the relative canonical equality to the surface resolution. All its exceptional discrepancy coefficients are positive. A normal surface germ with this property is smooth. Indeed, negative definiteness of the exceptional intersection matrix shows that a positive exceptional discrepancy divisor has negative intersection with some exceptional curve. That curve has negative canonical degree; adjunction makes it a smooth rational (-1) -curve. Contract it smoothly and repeat. The remaining discrepancy coefficients stay positive by pushforward. Eventually no exceptional curve remains, and the proper bimeromorphic map to the normal surface is an isomorphism.

On the other hand, a slice of a singular fourfold point by two equations has embedding dimension at least the ambient embedding dimension minus two, and hence greater than two. The chosen surface is singular there, a contradiction. $\square$

Lemma 4.5. *Let (T, Φ) be an ordinary klt pair on a compact normal complex fourfold, where $\Phi \geq 0$ is real and $a(E; T, \Phi) > 1$ for every exceptional prime divisor E over T . Apart from finitely many*

exceptional places, every exceptional place with $a(E; T, \Phi) < 2$ has centre a surface $S \subset T$ with the following properties: T and the reduced support of Φ are generically smooth along S , exactly one positive boundary component contains S , and, if its coefficient is b , then

$$a(E; T, \Phi) \geq 2 - b.$$

All places above such a surface are allowed in this conclusion, not just the exceptional divisor of its first blowup.

Proof. Choose a log resolution on which the positive strict boundary components are smooth and mutually disjoint. Starting with simple normal crossing support, blow up intersections of the positive components to separate them. Every exceptional crepant coefficient is negative by terminality. Thus the only positive components on this resolution are those disjoint strict transforms, with their original coefficients.

Consider a place not already on the resolution. At its general centre, apply (4.3), dropping negative boundary coefficients. A centre of codimension at least three has log discrepancy at least $3 - b > 2$, if it lies in a positive component of coefficient b , and at least three otherwise. A codimension-two centre outside the positive support has log discrepancy at least two. Thus a place with $a < 2$ has codimension-two centre in one positive component, and its log discrepancy is at least $2 - b$.

If this centre lies in the exceptional locus, it is an irreducible component of the intersection of that positive component and an exceptional divisor. There are finitely many such centres on the compact resolution. We check that each supports only finitely many places with $a < 2$. It suffices to retain only the smooth component of coefficient b , since deleting the negative coefficients lowers discrepancies. Make this reduction once on the fixed resolution; on subsequent blowups retain all exceptional coefficients in the crepant boundary. Blow up the fixed centre at its general point. Unless the given place has appeared, the same estimate forces its next centre to be the intersection of the positive strict transform and the new exceptional divisor. Indeed, the strict transform of the positive divisor maps isomorphically to that divisor near the generic point of the original centre, because the centre is Cartier within it. A codimension-two centre over that generic point must therefore be this unique intersection. A proper subset has codimension at least three and discrepancy at least $3 - b$; a centre outside the positive divisor has discrepancy at least two. Thus no other branch is possible. The process reaches the given place after finitely many blowups, by the decreasing positive integer empty-boundary discrepancy used in Lemma 4.2. For this reduced pair, the k th exceptional divisor on the branch has crepant coefficient $-k(1 - b)$ and log discrepancy

$$1 + k(1 - b), \quad k = 1, 2, \dots$$

Indeed, the first coefficient is $b - 1$, and blowing up the intersection of the newest exceptional divisor with the coefficient- b strict transform changes $-k(1 - b)$ to $b - k(1 - b) - 1 = -(k + 1)(1 - b)$. Thus only $k < 1/(1 - b)$ can occur before the cutoff two is reached. This proves finiteness over every fixed exceptional centre. These generic blowups are realized globally by blowing up the closures of the centres.

Add the finitely many exceptional primes on the chosen resolution. All other centres under consideration lie outside its exceptional locus at their general points, where the resolution is an isomorphism. Their images are precisely the surfaces described in the statement. $\square$

We now have the two counts needed for termination. A flipped surface will give a place whose discrepancy strictly increases into a fixed finite set. Once such surfaces disappear, a compact surface-class rank will decrease at every remaining step.

4.4. Real terminal pairs.

Proposition 4.6. *Let (T_0, Φ_0) be an ordinary klt pair on a normal globally $\mathbb{Q}$ -factorial compact Kähler fourfold, with K_{T_0} a $\mathbb{Q}$ -line bundle and effective real boundary. Suppose $a(E; T_0, \Phi_0) > 1$ for*

every exceptional prime divisor E over T_0 . Every sequence of ordinary $(K_{T_i} + \Phi_i)$ -flips starting at this pair terminates, provided all models are normal globally $\mathbb{Q}$ -factorial compact Kähler fourfolds, each K_{T_i} is a $\mathbb{Q}$ -line bundle, and each flip is given by projective small contractions

$$T_i \xrightarrow{f_i} Z_i \xleftarrow{f_i^+} T_{i+1}$$

with $-(K_{T_i} + \Phi_i)$ relatively ample for f_i and $K_{T_{i+1}} + \Phi_{i+1}$ relatively ample for f_i^+ . Here Φ_{i+1} is the strict transform of Φ_i . No pseudo-effectivity or bigness assumption is required.

Proof. By Lemma 2.3, discrepancies do not decrease, and they increase strictly for a place centred in either exceptional locus. Since the maps are small, the exceptional places are the same on all models. Thus every pair remains terminal in the stated sense, and its ambient fourfold is smooth in codimension two by Lemma 4.4.

Write the distinct positive coefficients of the boundary in decreasing order. We successively remove, by discarding finitely many initial steps, every flipping and flipped surface contained in a component of each coefficient. After the positive coefficients we treat the value zero, at which stage every flipped surface is considered.

Eliminating flipped surfaces. Fix the current coefficient b , and suppose that no flipping or flipped surface is contained in a component of coefficient greater than b . For $b = 0$ assume that all positive coefficients have already been treated. If S is a flipped surface contained in a coefficient- b component, or any flipped surface when $b = 0$, blow it up at its general point. The exceptional prime E has new log discrepancy

$$a(E; T_{i+1}, \Phi_{i+1}) = 2 - \sum_{l=1}^s b_l \text{mult}_S \Phi_{l,i+1}, \quad 1 < a(E; T_{i+1}, \Phi_{i+1}) \leq 2 - b, \quad (4.7)$$

where $\Phi_{l,i+1}$ are the positive prime boundary components and b_l their fixed coefficients. The blowup place is defined globally by blowing up the closed surface and resolving; its generic point suffices for this computation.

The values in (4.7) belong to a finite set, even for irrational coefficients. Indeed, terminality gives

$$\sum_l b_l \text{mult}_S \Phi_{l,i+1} < 1, \quad 0 \leq \text{mult}_S \Phi_{l,i+1} < 1/b_l.$$

Each multiplicity is an integer, and there are finitely many positive components. We use no discreteness assertion for all discrepancies.

Strictness in Lemma 2.3 gives $a(E; T_i, \Phi_i) < 2 - b$. Its discrepancy on the first model of the current tail is therefore also strictly below $2 - b$. By Lemma 4.5, outside a finite set such a place would initially have a surface centre in one boundary component of coefficient c , with discrepancy at least $2 - c$. Necessarily $c > b$. That centre persists as a surface in this component at every later step: at each step the map is an isomorphism at its general point, since otherwise the centre itself would be a flipping surface in a higher-coefficient component. It cannot become a flipped surface either, by the same higher-coefficient exclusion. Thus it cannot be the place just constructed. All these witness places belong to a fixed finite set.

Each member of that finite set is used only finitely often. Between uses its discrepancy does not decrease, and at each use it strictly increases to one of the finitely many values (4.7). Consequently only finitely many steps have a flipped surface of the current kind. Pass beyond them.

Eliminating flipping surfaces inside the boundary. Suppose $b > 0$. Fix a coefficient- b component and normalize its transforms on the two sides of a flip. Denote the normalizations by D_i^ν, D_{i+1}^ν , and normalize their common image in Z_i to obtain B_i . The induced maps to B_i are proper and bimeromorphic. They are isomorphisms away from the inverse image of the common exceptional image in Z_i , whose dimension is at most one by Lemma 2.3. Finite normalization preserves this

bound. On D_{i+1}^ν its inverse image also has dimension at most one, because no flipped surface is contained in the boundary component.

Every compact surface in B_i has a strict transform on D_i^ν , so pushforward gives a surjection

$$\mathcal{C}_2(D_i^\nu) \longrightarrow \mathcal{C}_2(B_i).$$

On the other side, Borel–Moore localization along the exceptional subset of dimension at most one makes restriction in degree four injective. Compatibility with proper pushforward and the common isomorphic open therefore gives an injection

$$\mathcal{C}_2(D_{i+1}^\nu) \hookrightarrow \mathcal{C}_2(B_i).$$

These are assertions about surface spans; no surjectivity on the full fourth homology of the negative side is needed. Only compact analytic topology is used on the normalized components and their common image; the positivity needed next comes from the ambient Kähler fourfold.

If a flipping surface lies in this boundary component, a surface above it on D_i^ν maps to a set of dimension at most one in B_i . Its class is nonzero: its image in T_i is a surface, and the square of a Kähler class on T_i has strictly positive integral over it. One may compute this integral on a resolution of the surface. The finite normalization has positive generic degree on this surface, so the same nonvanishing holds upstairs. Its class is a nonzero kernel vector in the displayed surjection. Hence

$$c_2(D_i^\nu) \geq c_2(D_{i+1}^\nu),$$

with strict inequality whenever the component contains a flipping surface. Summing over the finitely many coefficient- b components shows that only finitely many such steps occur. This completes the induction at b .

The remaining flips. After the positive coefficients have been treated, perform the flipped-surface argument with $b = 0$. No flipped surface remains. For every remaining flip, Lemma 2.3 gives

$$\dim \text{Exc}(f_i) + \dim \text{Exc}(f_i^+) \geq 3.$$

Smallness bounds both dimensions by two, so $\text{Exc}(f_i)$ contains a surface and $\dim \text{Exc}(f_i^+) \leq 1$. Apply Lemma 4.1 with $k = 2$ to the common isomorphic open. It gives $c_2(T_i) > c_2(T_{i+1})$ at every remaining step, impossible for an infinite sequence of nonnegative integers. $\square$

The algebraic discrepancy and cycle-count strategy originates in [15, Theorem 5-1-15 and Lemmas 5-1-16–5-1-17] and [8, 9]. The proof above supplies the real-boundary argument and uses compact analytic cycle classes with Kähler positivity. It applies those geometric tools one step at a time, without a common projective base.

Corollary 4.7. *Let X_0 be a normal globally strongly $\mathbb{Q}$ -factorial compact Kähler fourfold with terminal singularities. Every sequence starting at X_0 of ordinary empty-boundary negative divisorial contractions and flips, with the projectivity and compact Kähler conclusions of Proposition 3.4, terminates. No pseudo-effectivity hypothesis is required for this assertion about birational sequences.*

Proof. For flips, terminality persists by Lemma 2.3. For a divisorial contraction, (3.5) gives $K_{X_i} = f^*K_{X_{i+1}} + eE$ with $e > 0$. The lost prime has target log discrepancy $1 + e > 1$; every other exceptional place has target discrepancy at least its source discrepancy, which was greater than one. Thus terminality persists throughout the sequence. Lemma 4.1 with $k = 3$ bounds the number of divisorial steps. Any infinite sequence would therefore have a tail consisting only of flips, contrary to Proposition 4.6 with zero boundary. $\square$

5. SCALING AND A LIMITING TERMINAL PAIR

We now construct a particular ordinary program on the pair of Theorem 1.2. If that program were infinite, we would obtain strongly $\mathbb{Q}$ -factorial auxiliary models carrying nef generalized adjoints and a decreasing sequence of effective boundaries. Their limit will be an ordinary terminal pair. These auxiliary extractions are used only to compare the original program with a terminating one; the original program starts on X .

Fix a smooth Kähler form h on X , with class H , large enough that $K_X + \Delta + H$ is nef. Throughout this section $\mathbf{H}$ denotes the fixed b-nef datum obtained by pulling back this very form. On subsequent models its trace class is denoted by H_T , as in Lemma 2.4. We do not replace the fixed datum by arbitrary cohomologous current representatives.

5.1. Constructing the scaling program.

Proposition 5.1. *Let (X, Δ) satisfy the hypotheses of Theorem 1.2. There is an ordinary negative-ray program, finite or infinite,*

$$(X, \Delta) = (X_0, \Delta_0) \dashrightarrow (X_1, \Delta_1) \dashrightarrow \cdots,$$

in the same global Weil-divisor $\mathbb{Q}$ -factorial compact Kähler category. It stops when the ordinary adjoint is nef or a negative ray has a Mori fibre contraction. Put

$$D_i = K_{X_i} + \Delta_i, \quad L_i(t) = D_i + tH_i, \quad H_i = H_{X_i}, \quad \lambda_{-1} = 1.$$

At every stage before stopping there is a parameter $0 < \lambda_i \leq \lambda_{i-1}$ and a contracted extremal ray R_i such that

$$D_i \cdot R_i < 0, \quad L_i(\lambda_i) \cdot R_i = 0, \quad L_i(\lambda_i) \text{ is nef.}$$

The generalized pairs $(X_i, \Delta_i + t\mathbf{H})$ are gklt for $0 \leq t \leq \lambda_{i-1}$. A birational step is crepant for the data at $t = \lambda_i$, and discrepancies do not decrease across that step for any $0 \leq t \leq \lambda_i$.

Proof. On X_0 , the nef datum descends, so its addition changes no discrepancies. Thus all the initial generalized pairs are gklt. Suppose the assertions have been established through X_i . If D_i is nef, stop. Otherwise define

$$\lambda_i = \min\{t \in [0, \lambda_{i-1}] : L_i(t) \text{ is nef}\}.$$

The set is nonempty by induction and is closed and convex. Its minimum is positive because D_i is not nef.

We first find a ray at this threshold. Choose $t_k \uparrow \lambda_i$ with $\lambda_i/2 < t_k < \lambda_i$. The generalized cone theorem gives an extremal ray negative for $L_i(t_k)$. Since $L_i(\lambda_i)$ is nef, that ray is also negative for $L_i(\lambda_i/2)$. The trace H_i is big by Corollary 2.5. Hence $\Delta_i + (\lambda_i/2)H_i$ is big, and the cone theorem gives only finitely many negative rays for this latter generalized pair [13, Theorem 1.3]. One ray R_i therefore occurs for infinitely many k . On this fixed ray, continuity gives $L_i(\lambda_i) \cdot R_i = 0$. Moreover $H_i \cdot R_i > 0$, so $D_i \cdot R_i < 0$.

Apply Proposition 3.4 to this ordinary negative ray, and denote its contraction by $f_i : X_i \rightarrow Z_i$. A fibre contraction gives the stopping alternative. Otherwise make the ordinary divisorial step or flip. Lemma 3.5 transports the trace H_i to H_{i+1} while preserving its relation to the fixed b-datum. By Lemma 2.1, the class $L_i(\lambda_i)$ descends to Z_i . Its descended class is nef, and its pullback on the new model is $L_{i+1}(\lambda_i)$. Thus the two threshold classes have equal pullbacks on a common resolution. Their difference is represented by the actual exceptional discrepancy divisor, which vanishes by negativity as in Lemma 2.6. The step is consequently generalized crepant at the threshold.

For completeness, this also proves the induction on singularities. On a common resolution, the log discrepancy at each prime E is affine in t . Its difference across the step is zero at $t = \lambda_i$ and is

the ordinary discrepancy increase at zero. Consequently

$$\begin{aligned} & a(E, X_{i+1}, \Delta_{i+1} + t\mathbf{H}) - a(E, X_i, \Delta_i + t\mathbf{H}) \\ &= \left(1 - \frac{t}{\lambda_i}\right) (a(E, X_{i+1}, \Delta_{i+1}) - a(E, X_i, \Delta_i)) \geq 0 \end{aligned} \quad (5.1)$$

for $0 \leq t \leq \lambda_i$. The generalized pairs on this interval remain gklt, and the new upper-threshold class is nef. The ordinary steps preserve all the stated properties of the original category by Proposition 3.4. $\square$

5.2. An infinite program has vanishing thresholds. We record first why an infinite program has a tail consisting only of flips. Use the cycle-space notation of Lemma 4.1: $c_3(T)$ is the dimension of the span of compact analytic three-dimensional cycle classes in $H_6(T, \mathbb{R})$. It is finite. A small modification of fourfolds is an isomorphism away from sets of dimension at most two, so that lemma, applied in both directions, preserves c_3 . A divisorial contraction strictly lowers it: an exceptional divisor has a nonzero cycle class, detected by integration of the cube of a Kähler form, and maps to a set of dimension at most two. Therefore only finitely many divisorial steps occur.

Lemma 5.2. *If the program of Proposition 5.1 is infinite, then $\lambda_i \rightarrow 0$.*

Proof. Fix $0 < \epsilon < 1$ and a resolution $\nu : U \rightarrow X$. Consider the compact segment of generalized data

$$\mathcal{P}_\epsilon = \{(\Delta, t\nu^*H) : \epsilon \leq t \leq 1\} \subset \text{Div}_{\mathbb{R}}(X) \times H_{\text{BC}}^{1,1}(U).$$

The upstairs nef data here are the pullbacks of the fixed smooth form. Every pair in this segment is gklt, its upstairs class is nef, and its boundary-plus-trace is modified big, since tH is Kähler on X . This real segment is allowed in the finiteness theorem for weak log canonical models [13, Theorem 5.15]; neither rationality of H nor strong $\mathbb{Q}$ -factoriality of X is required there.

Let $\phi_i : X \dashrightarrow X_i$ be the accumulated map, and suppose $\lambda_i \geq \epsilon$. It is a weak log canonical model of the original generalized pair at parameter λ_i . Indeed, it extracts no divisors, its target adjoint is nef, and discrepancies do not decrease: apply (5.1) to every preceding step at the fixed parameter λ_i , which is no larger than any preceding threshold. On a common resolution the original adjoint is the pullback of the target nef class plus an effective exceptional divisor. Pseudo-effectivity descends by Lemma 2.2, so the original data belong to the pseudo-effective portion of $\mathcal{P}_\epsilon$.

The finiteness conclusion concerns the maps ϕ_i , up to isomorphisms of their targets commuting with the maps from X . These marked maps are pairwise distinct in the flip tail. In fact, given two stages, an exceptional prime has strictly increasing ordinary discrepancy at the first intervening flip by Lemma 2.3, and later steps cannot decrease it. A target isomorphism compatible with the marking would identify the transformed boundary and all its discrepancies, a contradiction. This is also the distinction used in the proof of [13, Theorem 5.16].

There are therefore only finitely many i with $\lambda_i \geq \epsilon$. Since $\epsilon > 0$ was arbitrary, the thresholds tend to zero. $\square$

An infinite program would thus provide nef generalized adjoints with arbitrarily small nef parts. We next put their nonterminal places on auxiliary models, so that the boundary can be allowed to converge on one fixed space.

5.3. Stabilizing the extractions.

Proposition 5.3. *Suppose the program of Proposition 5.1 is infinite. After discarding finitely many stages, there are proper bimeromorphic morphisms*

$$\mu_i : Y_i \longrightarrow X_i$$

with Y_i strongly $\mathbb{Q}$ -factorial and compact Kähler, and effective real boundaries Θ_i , with the following properties.

(i) The natural maps between the Y_i are isomorphisms in codimension one, and

$$Q_{i,Y_i} := K_{Y_i} + \Theta_i + \lambda_i H_{Y_i} = \mu_i^* L_i(\lambda_i) \quad (5.2)$$

is nef. Each ordinary pair (Y_i, Θ_i) is klt and terminal.

(ii) On a fixed first model $Y = Y_{i_0}$, write $\Theta_{i,Y}$ for the strict transform of Θ_i . These boundaries have a fixed finite containing support and decrease coefficientwise to an effective real boundary Θ_Y . The ordinary pair (Y, Θ_Y) is klt and terminal.

(iii) The strict transform of Θ_Y on each Y_i pushes forward to Δ_i under μ_i .

Here terminal means that every exceptional prime has log discrepancy strictly greater than one.

Proof. Discard the finitely many divisorial steps. All remaining maps $X_i \dashrightarrow X_{i+1}$ are small. For each such i , put

$$\mathcal{E}_i = \{E \text{ exceptional over } X_i : a(E, X_i, \Delta_i + \lambda_i \mathbf{H}) \leq 1\}.$$

The fixed nef datum has a carrier projective over X_i : take a common resolution of the finitely many preceding projective ordinary step diagrams. For a flip its graph is projective over either side, so this construction and a further projective log resolution retain projectivity over X_i . On this resolution the datum is the pullback of h . Lemma 4.2 therefore makes $\mathcal{E}_i$ finite. It allows real coefficients and counts places of log discrepancy exactly one as well as those below one.

Smallness identifies the primes exceptional over successive X_i . Their log discrepancies at the current threshold agree by crepancy. On the new model, decreasing the coefficient of the nef part increases discrepancies: in the trace formula of Lemma 2.4, the correction divisor is effective. Thus, for every such prime,

$$\begin{aligned} a(E, X_{i+1}, \Delta_{i+1} + \lambda_{i+1} \mathbf{H}) &\geq a(E, X_{i+1}, \Delta_{i+1} + \lambda_i \mathbf{H}) \\ &= a(E, X_i, \Delta_i + \lambda_i \mathbf{H}). \end{aligned} \quad (5.3)$$

It follows that $\mathcal{E}_{i+1} \subseteq \mathcal{E}_i$. After another truncation they equal one finite set $\mathcal{E}$. Only this set stabilizes; no discreteness of the real discrepancy values is asserted.

Apply the exact extraction theorem [13, Corollary 2.29] to the generalized pair at λ_i and the prescribed set $\mathcal{E}$. It gives a strongly $\mathbb{Q}$ -factorial compact Kähler model $\mu_i : Y_i \rightarrow X_i$ extracting precisely those primes. Its compact Kähler output is supplied by the relative MMP in its proof. The input of this theorem need not be strongly $\mathbb{Q}$ -factorial, and its nef datum and crepant boundary may be real.

Let Θ_i be the crepant generalized boundary on Y_i . On the transform of a divisor of X_i , its coefficient is the coefficient in Δ_i ; on an extracted prime E , it is

$$1 - a(E, X_i, \Delta_i + \lambda_i \mathbf{H}) \in [0, 1).$$

Thus Θ_i is effective. The generalized crepant relation gives (5.2). More explicitly, subtract the ordinary class $K_{Y_i} + \Theta_i$ from the pulled-back generalized adjoint and divide by $\lambda_i > 0$. Lemma 2.4 identifies the result as the trace H_{Y_i} of our fixed b-datum, with an effective exceptional correction on a common resolution. Equation (5.2) is nef because its right-hand side is a pullback of a nef class.

Every prime exceptional over Y_i is exceptional over X_i and is absent from $\mathcal{E}$, so its generalized log discrepancy is greater than one. Dropping the nef part weakly increases this discrepancy. Hence (Y_i, Θ_i) is ordinary terminal. Together with the boundary coefficients below one, this also proves that it is klt.

The flip tail identifies all prime divisors on the X_i , and each Y_i extracts exactly the same additional primes. Consequently the Y_i are naturally isomorphic in codimension one. Fix the first one, $Y = Y_{i_0}$. On its primes inherited from X_{i_0} , the coefficients of $\Theta_{i,Y}$ are constant. On the extracted primes,

they decrease by (5.3). They are nonnegative and have a fixed finite containing support, so their coefficientwise limit Θ_Y exists and

$$0 \leq \Theta_Y \leq \Theta_{i,Y} \leq \Theta_{i_0,Y}.$$

Terminality of the limit follows by comparison with the first boundary on this fixed model. Since Y is $\mathbb{Q}$ -factorial, the effective real divisor $\Theta_{i_0,Y} - \Theta_Y$ has nonnegative order at every divisorial valuation. Therefore

$$a(E, Y, \Theta_Y) \geq a(E, Y, \Theta_{i_0,Y}) > 1$$

for every exceptional prime E . All boundary coefficients remain below one, so the limit is klt as well. This argument uses decrease of the boundary, rather than any assertion that terminality is closed under limits. Finally, only coefficients of μ_i -exceptional primes vary. The transformed limiting boundary thus pushes forward to Δ_i , proving (iii). $\square$

The infinite-program assumption has now produced an ordinary terminal pair on a fixed strongly $\mathbb{Q}$ -factorial model, together with the nef adjoints (5.2). In the next section we apply terminal termination to that limiting pair and compare a small perturbation with one of these nef adjoints.

6. THE LIMITING MODEL AND THE PERTURBATION CONTRADICTION

We now show that the ordinary program constructed in Proposition 5.1 is finite. An infinite program would give, by Proposition 5.3, a decreasing family of boundaries on small modifications of one terminal pair. We first make the limiting ordinary adjoint nef. We then compare it with one sufficiently nearby generalized adjoint. The decisive point is that this comparison can be made without changing the Cartier indices of finitely many rational nef adjoints. The rational decomposition and perturbation method follows Birkar's argument [1, Remark 3.1, Lemma 3.2, and Section 4]. We give the details needed for the fixed Kähler b-part and for exact line-bundle descent on the analytic contractions.

6.1. A nef model of the limiting ordinary pair. Recall the data supplied by Proposition 5.3 under the assumption that the scaling program is infinite. After truncation, all its steps are flips. There are strongly $\mathbb{Q}$ -factorial compact Kähler models $\mu_i : Y_i \rightarrow X_i$, isomorphic to one another in codimension one, and effective boundaries Θ_i such that

$$Q_{i,Y_i} := \mu_i^* L_i(\lambda_i) = K_{Y_i} + \Theta_i + \lambda_i H_{Y_i} \quad \text{is nef.} \quad (6.1)$$

Here $L_i(t) = D_i + tH_{X_i}$, $D_i = K_{X_i} + \Delta_i$, and $\lambda_i \rightarrow 0$. The nef b-part $\mathbf{H}$ is the fixed datum determined by the initial Kähler form. On a fixed first model Y , the transforms $\Theta_{i,Y}$ decrease coefficientwise, on a fixed finite support, to an effective real boundary Θ_Y . The ordinary pair (Y, Θ_Y) is terminal. The transform of Θ_Y on each Y_i pushes forward to Δ_i .

Proposition 6.1. *For these data there is a finite sequence of ordinary $(K_Y + \Theta_Y)$ -flips*

$$Y \dashrightarrow W$$

such that W is strongly $\mathbb{Q}$ -factorial and compact Kähler, the transformed ordinary pair (W, Θ_W) is terminal, and $P := K_W + \Theta_W$ is nef. In particular, W is isomorphic in codimension one to every Y_i .

Proof. Run the ordinary real-boundary program from (Y, Θ_Y) using [13, Proposition 4.2], with zero nef part. We show inductively that every step is a flip. Let T be a current model, so that T is small to every Y_i , and write

$$P_T = K_T + \Theta_T, \quad Q_{i,T} = K_T + \Theta_{i,T} + \lambda_i H_T.$$

The trace H_T exists by Lemma 2.4: along these strongly $\mathbb{Q}$ -factorial real-boundary flips we use the canonical Bott–Chern transport of [13, Proposition 4.2]. Its pullback comparison has an actual exceptional-divisor correction, so the same trace relation propagates with the fixed b-data. Together

with the effective transformed boundary, the actual trace relation supplies the generalized structure boundary for $Q_{i,T}$; no generalized klt condition is required for the small-model comparison below. The finite boundary support and $\lambda_i \rightarrow 0$ give $Q_{i,T} \rightarrow P_T$ in $H_{\text{BC}}^{1,1}(T)$.

Fix a P_T -negative extremal ray and its contraction. For all sufficiently large i this same ray is $Q_{i,T}$ -negative. Take a common smooth Kähler resolution $p : U \rightarrow T$, $q : U \rightarrow Y_i$. The boundaries are transforms of one another and the b-part is unchanged. Thus Lemma 2.6, together with (6.1), gives an actual divisor E_i such that

$$p^*Q_{i,T} = q^*Q_{i,Y_i} + [E_i], \quad E_i \geq 0, \quad E_i \text{ is exceptional over both models.} \quad (6.2)$$

Suppose that the contraction were divisorial. Choose a general point of an exceptional prime outside the codimension-at-least-two image of $\text{Exc}(p)$. A positive-dimensional projective fibre through this point contains an irreducible curve C . Its strict transform $\tilde{C}$ on U is not contained in $\text{Supp}(E_i)$. Consequently

$$Q_{i,T} \cdot C = q^*Q_{i,Y_i} \cdot \tilde{C} + E_i \cdot \tilde{C} \geq 0,$$

contrary to the choice of the ray. For a fibre-type contraction the same argument uses a general point of T instead. It also rules out ending the program with a Mori fibre space.

Therefore every step is an ordinary flip. Terminality persists by the ordinary discrepancy comparison. These flips have the exact relative properties needed for the auxiliary termination theorem. The negative contraction is projective by [13, Proposition 4.2]; the positive one is proper birational between rational-singularity spaces, hence strongly Moishezon by [13, Lemma 2.39], and projective by [13, Lemma 2.40]. For a step $T \xrightarrow{f} Z \xleftarrow{f^+} T^+$, the signed ordinary adjoints $-P_T$ and P_{T^+} are relatively Kähler real line bundles. To obtain relative ampleness in this intrinsic line-bundle setting, write either signed adjoint L in a finite real span of integral line bundles, and choose a base class γ such that $c_1(L) + h^*\gamma$ is Kähler, where $h = f$ or f^+ . Openness of the Kähler cone gives a rational simplex around the coefficient vector of L whose rational vertices L_ν still have $c_1(L_\nu) + h^*\gamma$ Kähler. Lemma 3.3 makes every L_ν relatively ample, and their positive convex combination L is therefore relatively ample as a real line bundle. Thus Proposition 4.6 makes this sequence finite. Its endpoint has nef ordinary adjoint, as asserted. $\square$

6.2. Rational nef boundaries and a positive intersection gap. The following argument gives a rational description near a nef ordinary adjoint, even though the limiting boundary has real coefficients. Only the ordinary boundary coefficients vary in this argument; the transcendental b-part plays no role.

Lemma 6.2. *Let W be a globally $\mathbb{Q}$ -factorial compact Kähler fourfold with K_W a $\mathbb{Q}$ -line bundle. Let $B \geq 0$ be a real boundary such that (W, B) is klt and $K_W + B$ is nef. There are effective rational klt boundaries $B^1, \dots, B^r$ and positive real numbers $u_1, \dots, u_r$ such that*

$$\sum_{j=1}^r u_j = 1, \quad B = \sum_{j=1}^r u_j B^j, \quad K_W + B^j \text{ is nef for every } j.$$

The boundaries B^j can be chosen on the positive support of B .

Proof. If $B = 0$, take $r = 1$ and $B^1 = 0$. Otherwise work in the finite-dimensional coefficient space on $\text{Supp}(B)$. A sufficiently small simplex with rational vertices $C^0, \dots, C^d$ contains B in its interior and consists of effective klt boundaries. Indeed, all coefficients of B on this support are positive, and the klt inequalities are open on one log resolution. Put $D_k = K_W + C^k$.

Choose a rational number $\delta > 0$ smaller than all the barycentric coordinates of B . Let $\mathcal{P}$ be the smaller rational polytope in the simplex defined by requiring every barycentric coordinate to be at least δ . We claim that its nef locus

$$\mathcal{N} = \{C \in \mathcal{P} : K_W + C \text{ is nef}\}$$

is a rational polytope.

Consider an extremal ray R that is negative for $K_W + C$ for some $C \in \mathcal{P}$. At least one vertex adjoint, say D_a , is negative on R . Among all irreducible rational curves whose classes span R , choose Γ minimizing $-D_a \cdot \Gamma$. Such curves exist by the cone theorem [13, Theorem 1.3], and a minimum exists because a Cartier multiple of D_a puts their positive degrees in a discrete subset of $\mathbb{R}_{>0}$.

This choice minimizes the numerical scale of the curve class on R . For any other vertex D_k negative on R , the cone theorem supplies a rational curve Γ_k on R with $0 < -D_k \cdot \Gamma_k \leq 8$. Since $[\Gamma]$ is a positive multiple of $[\Gamma_k]$ with multiplier at most one, it follows that $-D_k \cdot \Gamma \leq 8$ as well. Vertices nonnegative on R need no estimate. Thus all the numbers $d_k = D_k \cdot \Gamma$ satisfy $d_k \geq -8$.

Write a member of $\mathcal{P}$ that is negative on R as $\sum t_k C^k$, where $t_k \geq \delta$ and $\sum t_k = 1$. Then $\sum t_k d_k < 0$, so for each k ,

$$t_k d_k < 8(1 - t_k).$$

In particular,

$$-8 \leq d_k < \frac{8}{\delta} \quad (0 \leq k \leq d). \quad (6.3)$$

Each D_k has a fixed global Cartier index on W . Hence its degrees belong to a fixed discrete subgroup of $\mathbb{R}$, and (6.3) permits only finitely many vectors $(d_0, \dots, d_d)$. Their nonnegativity inequalities are rational.

These finitely many inequalities define exactly $\mathcal{N}$ inside $\mathcal{P}$. Indeed, a ray never negative on $\mathcal{P}$ imposes no further condition. For any other ray the chosen vector tests its nonnegativity. Finally, positivity on all extremal rays is positivity on $\overline{\text{NA}}(W)$: intersect this cone with the hyperplane where a fixed Kähler class has value one and use the resulting compact convex slice. Nef-cone duality now proves the claim. Express B as a convex combination of rational vertices of $\mathcal{N}$, omitting vertices with zero weight. These are the required B^j . $\square$

Apply Lemma 6.2 to (W, Θ_W) from Proposition 6.1. Fix a decomposition

$$\Theta_W = \sum_{j=1}^r u_j B^j, \quad P = \sum_{j=1}^r u_j P^j, \quad P^j = K_W + B^j, \quad (6.4)$$

and positive integers ℓ_j such that $\ell_j P^j$ is an integral line bundle, also clearing the coefficients of B^j . Define

$$c = \min_{1 \leq j \leq r} \frac{u_j}{\ell_j} > 0, \quad m \in \mathbb{N}, \quad (m-1)c > 8. \quad (6.5)$$

Whenever the transforms of the P^j remain nef and the same ℓ_j remain Cartier indices, every curve Γ satisfies

$$P \cdot \Gamma > 0 \implies P \cdot \Gamma \geq c. \quad (6.6)$$

At least one nonnegative summand in (6.4) is then positive, and its degree is at least $1/\ell_j$. The next argument preserves exactly these indices.

6.3. A nearby program that is trivial for the limiting class.

Proposition 6.3. *For the extracted data of Proposition 5.3 and the model W of Proposition 6.1, take i sufficiently large. There is a finite generalized $Q_{i,W}$ -program $W \dashrightarrow W_i$ consisting only of flips, with the following properties:*

- (1) Q_{i,W_i} is nef;
- (2) every step is trivial for the transformed class P and for each P^j in (6.4);
- (3) the transformed P^j are nef adjoints of ordinary rational klt pairs, and the fixed multiples $\ell_j P^j$ are line bundles;
- (4) every Q_{i,W_i} -trivial curve is P_{W_i} -trivial.

Proof. On W , set $G_i = \Theta_{i,W} - \Theta_W \geq 0$. Keep the integer m in (6.5) fixed and consider the generalized adjoint

$$A_W = K_W + \Theta_W + mG_i + m\lambda_i H_W = mQ_{i,W} - (m-1)P. \quad (6.7)$$

For large i its pair is gklt. To see this, choose one log resolution carrying the finite boundary support and the fixed b-data. The crepant boundary coefficients vary continuously with G_i and λ_i ; at the limit they are those of the ordinary klt pair (W, Θ_W) . Thus they remain strictly less than one. The boundary $\Theta_W + mG_i$ is effective. The pair for $Q_{i,W}$ is the convex interpolation with weight $1/m$ between this pair and (W, Θ_W) , so it is gklt too.

The sum $\Theta_{i,W} + \lambda_i H_W$ is big: $\lambda_i > 0$, the trace H_W is big by Corollary 2.5, and $\Theta_{i,W}$ is effective. The model W is strongly $\mathbb{Q}$ -factorial. Choose a sufficiently large Kähler scaling class and apply [13, Theorem 1.5] to obtain a finite $Q_{i,W}$ -program.

All its steps are flips and its endpoint is nef. Indeed, as long as the current model T is small to Y_i , the comparison

$$p^*Q_{i,T} = q^*Q_{i,Y_i} + [E], \quad E \geq 0$$

holds on a common resolution, with E exceptional over both models. The curve-through-a-general-point argument in Proposition 6.1 excludes a negative divisorial contraction or a fibre-type contraction. It applies with this fixed i , without any limiting argument.

We prove the remaining assertions by induction along these flips. Suppose that on the current model T the ordinary pairs (T, B_T^j) are klt, the $P_T^j = K_T + B_T^j$ are nef with $\ell_j P_T^j$ line bundles, and the amplified pair with class

$$A_T = mQ_{i,T} - (m-1)P_T$$

is gklt. These conditions hold initially. A $Q_{i,T}$ -negative extremal ray is A_T -negative because P_T is nef. The cone theorem for the amplified pair gives a rational curve Γ spanning that ray with $A_T \cdot \Gamma \geq -8$. If $P_T \cdot \Gamma > 0$, then (6.6) implies

$$mQ_{i,T} \cdot \Gamma = A_T \cdot \Gamma + (m-1)P_T \cdot \Gamma \geq -8 + (m-1)c > 0,$$

a contradiction. Hence the ray is P_T -trivial. Since every u_j is positive and every P_T^j is nef, it is also P_T^j -trivial for each j .

Let $f : T \rightarrow Z \leftarrow T^+ : f^+$ be this flip. For the ordinary rational klt pair (T, B_T^j) , the line bundle $M_j = \ell_j P_T^j$ is numerically trivial over Z , and

$$M_j - (K_T + B_T^j) = (\ell_j - 1)P_T^j$$

is relatively nef. Lemma 3.1 gives an actual line bundle $M_{j,Z}$ with $M_j = f^*M_{j,Z}$. The pullback $(f^+)^*M_{j,Z}$ agrees on the common big open set with the reflexive sheaf of $\ell_j(K_{T^+} + B_{T^+}^j)$. Normality and reflexivity extend the agreement. Thus the *same* integer ℓ_j is a Cartier index on T^+ ; no further multiple is taken.

It follows that both adjoints P_T^j and $P_{T^+}^j$ pull back from the same class on Z . On a common smooth compact Kähler resolution $p : V \rightarrow T$, $q : V \rightarrow T^+$, their pullbacks are therefore equal. The class $p^*P_T^j$ is nef, and nef descent along q in Lemma 2.1 shows that $P_{T^+}^j$ is nef: the target T^+ is generalized klt and hence has rational singularities. Their actual discrepancy-change divisor is zero by negativity, so the pairs remain klt and the comparison is crepant. In particular the same is true of their convex combination P_T .

Finally, on a common resolution the discrepancy-change divisor for the amplified pair is now m times that for the Q_i -pair, because the contribution from P_T is zero. The latter divisor is effective along the chosen generalized flip. The amplified pair therefore remains gklt. This completes the induction, including the fixed-index assertion and the validity of (6.6) at the endpoint.

It remains to prove assertion (4). Write $Q = Q_{i,W_i}$ and $P' = P_{W_i}$. Both are nef. Suppose a Q -trivial curve has positive P' -degree. The face

$$\mathcal{F} = \{\gamma \in \overline{\text{NA}}(W_i) : Q \cdot \gamma = 0\}$$

then has a compact Kähler slice on which P' takes a positive value. Maximize P' on that slice and choose an extreme point of its maximizing face. This point is extreme in the slice of $\mathcal{F}$. Since $\mathcal{F}$ is a face of $\overline{\text{NA}}(W_i)$, it spans an extremal ray of the full cone. On this ray $Q = 0$ and $P' > 0$, so the amplified class $A = mQ - (m-1)P'$ is negative. Its cone theorem gives a rational generator Γ with

$$-8 \leq A \cdot \Gamma = -(m-1)P' \cdot \Gamma \leq -(m-1)c < -8,$$

which is impossible. $\square$

6.4. Descent to the original flipping base. The preceding construction has produced a nef limiting class that vanishes on every curve killed by the nearby nef generalized class. We now descend it to the base of one of the original flips. This forces the original negative adjoint to have degree zero on its chosen ray, giving the contradiction.

Proposition 6.4. *Let (X, Δ) satisfy the hypotheses of Proposition 5.1. The ordinary program constructed there, with scaling of the fixed initial Kähler b-part, is finite.*

Proof. Suppose it were infinite. Use Proposition 5.3 and choose W , then a sufficiently large i and W_i , as in Propositions 6.1 and 6.3. Let $f_i : X_i \rightarrow Z_i$ be the contraction of the chosen D_i -negative ray in the original flip tail. The threshold class $L_i(\lambda_i)$ descends to Z_i .

Take a common smooth compact Kähler resolution $p : V \rightarrow W_i$ and $q : V \rightarrow Y_i$. Put $s = \mu_i q$ and $r = f_i s$. The relevant maps are shown in Figure 2.

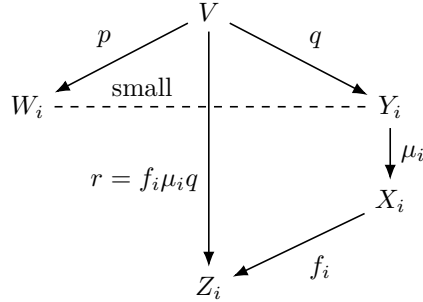


FIGURE 2. The final comparison is made on V . The class $p^*P_{W_i}$ descends along r ; no morphism from W_i to X_i or to Z_i is required.

The models W_i and Y_i are small modifications with the same transformed boundary and b-data. Their Q_i -classes are both nef. Applying Lemma 2.6 in both directions gives

$$p^*Q_{i,W_i} = q^*Q_{i,Y_i} = s^*L_i(\lambda_i). \quad (6.8)$$

Consequently this class has zero degree on every r -vertical curve. For such a curve C , if $p(C)$ is a point then $p^*P_{W_i} \cdot C = 0$ immediately. Otherwise the projection formula and (6.8) show that $p(C)$ is Q_{i,W_i} -trivial. Proposition 6.3(4) then gives $p^*P_{W_i} \cdot C = 0$ as well.

The map r is proper and birational. Its target Z_i is normal compact Kähler with rational singularities by the ordinary-step construction, and its source V is smooth compact Kähler. Its fibres are connected, and its general fibre is a point. Thus all the hypotheses of Lemma 2.1 apply to r . There is a class $\xi \in H_{\text{BC}}^{1,1}(Z_i)$ with

$$p^*P_{W_i} = r^*\xi. \quad (6.9)$$

We next form the ordinary canonical-and-boundary comparison. Using compatible canonical data, the difference

$$F = p^*(K_{W_i} + \Theta_{W_i}) - s^*(K_{X_i} + \Delta_i) \quad (6.10)$$

is an actual real divisor on V , with the relative canonical terms interpreted as in Section 2. It is exceptional over X_i : on every prime that is not s -exceptional the two boundary coefficients agree, since the limiting boundary on Y_i pushes to Δ_i and W_i is small to Y_i . This constructs the divisor before any appeal to its cohomology class.

By (6.9),

$$[F] = p^*P_{W_i} - s^*D_i = s^*(f_i^*\xi - D_i).$$

Hence F is numerically trivial over X_i . Lemma 2.3, applied to F and to $-F$, gives $F = 0$. Pullback injectivity for s now yields

$$D_i = f_i^*\xi \quad \text{in } H_{\text{BC}}^{1,1}(X_i).$$

Its intersection with the ray contracted by f_i is therefore zero, contradicting the strict D_i -negativity with which that ray was chosen. The scaling program is finite. $\square$

7. THE TWO ENDPOINTS

The scaling program is finite by Proposition 6.4. We now identify its endpoint from the pseudo-effectivity of the original adjoint. Only ordinary birational comparisons are needed for this last step.

Lemma 7.1. *Let $(T, B) \dashrightarrow (T', B')$ be an ordinary negative divisorial contraction or flip between normal compact Kähler klt pairs, with rational boundaries and rational line-bundle adjoints. Then $K_T + B$ is pseudo-effective if and only if $K_{T'} + B'$ is pseudo-effective.*

Proof. Set $D = K_T + B$ and $D' = K_{T'} + B'$. The ordinary comparison in Proposition 3.4 and Lemma 2.3 gives, on a common smooth compact Kähler resolution $p : V \rightarrow T$, $q : V \rightarrow T'$,

$$p^*D = q^*D' + F, \quad F \geq 0,$$

where F is an actual divisor exceptional over T' . If D' is pseudo-effective, its pullback plus $[F]$ is pseudo-effective; descent along p shows that D is pseudo-effective. Conversely, if D is pseudo-effective, exceptional descent along q applied to $q^*D' + [F]$ proves that D' is pseudo-effective. Both uses of pullback and descent are justified by Lemma 2.2. $\square$

Lemma 7.2. *Let $g : Y \rightarrow S$ be a projective surjective morphism of normal compact Kähler varieties with $\dim S < \dim Y$. If a rational line bundle D satisfies $-D$ g -ample, then D is not pseudo-effective.*

Proof. Suppose that $c_1(D)$ contains a closed positive current T with local plurisubharmonic potentials. Choose a point y where such a potential is finite and which lies on a positive-dimensional component of a fibre of g . This is possible on the dense open locus of fibres of the expected dimension: the pole set of a plurisubharmonic potential contains no open subset. That fibre component is projective, so hyperplane sections through y give an integral curve C through y contained in the fibre. Let $\nu : C^\nu \rightarrow Y$ denote its normalization followed by inclusion.

Local potentials pull back under ν to subharmonic functions or to the constant $-\infty$. Finiteness at a preimage of y excludes the latter alternative on the connected curve. Thus they define a positive current ν^*T in the class $\nu^*c_1(D)$, and

$$D \cdot C = \int_{C^\nu} \nu^*T \geq 0.$$

This contradicts $D \cdot C < 0$, which follows from relative ampleness. The choice of y is essential: no restriction of T to a curve contained in its pole set has been used. $\square$

Proof of Theorem 1.2. Run the scaling construction on the given pair (X, Δ) . Proposition 3.4 supplies each ordinary negative step, and Proposition 6.4 proves that the construction stops after finitely many such steps. Write its endpoint as (Y, Γ) . All intermediate varieties are normal compact Kähler fourfolds, all boundaries are the rational transforms of Δ , and all pairs remain ordinary klt and globally Weil-divisor $\mathbb{Q}$ -factorial with the stipulated canonical datum. The optional strong global property is preserved by the same Proposition. The auxiliary extractions used to prove finiteness do not change this program, which starts on X itself.

The stopping rule gives either a nef adjoint $K_Y + \Gamma$ or a Mori contraction $g : Y \rightarrow S$. In the latter case Proposition 3.4 supplies a normal compact Kähler base, a projective surjective morphism with connected fibres, $\dim S < 4$, relative numerical rank $\rho(Y/S) = 1$, and relative ampleness of $-(K_Y + \Gamma)$. These are precisely the required Mori fibre space conditions, with relative rank computed from global Cartier classes. The same Proposition gives relative Bott–Chern dimension one for every negative contraction, including g .

By Lemma 7.1, the adjoint at every stage has the same pseudo-effectivity status as $K_X + \Delta$. If the latter is pseudo-effective, Lemma 7.2 excludes the Mori alternative, so the endpoint is nef. If it is not pseudo-effective, the endpoint cannot be nef, since nef classes are pseudo-effective by Lemma 2.2; hence the endpoint has the asserted Mori fibre structure.

For completeness, the nef endpoint has the discrepancy properties of a log minimal model. No step extracts a divisor. On a common resolution of the finite sequence, with maps $p : V \rightarrow X$ and $q : V \rightarrow Y$, the ordinary effective comparison divisors add to

$$p^*(K_X + \Delta) = q^*(K_Y + \Gamma) + F, \quad F \geq 0,$$

with F exceptional over Y . Consequently, for every prime divisor E over these models,

$$a(E, Y, \Gamma) \geq a(E, X, \Delta).$$

If a prime divisor on X is contracted, its discrepancy increases strictly at the divisorial step that contracts its transform and never decreases thereafter. Thus the displayed inequality is strict for every such prime. Here a denotes the log discrepancy, as throughout the paper.

If the initial adjoint is nef, the construction takes no steps; conversely, a zero-step nef outcome requires the initial adjoint to be nef. In the non-pseudo-effective case a zero-step program is also allowed when the first selected contraction already gives a Mori fibre structure on X . This completes both endpoint assertions. $\square$

APPENDIX A. WHY ANALYTIC LOCAL FACTORIALITY IS DIFFERENT

Global Weil-divisor $\mathbb{Q}$ -factoriality does not require every analytic local ring to be $\mathbb{Q}$ -factorial. The distinction already occurs for projective threefolds with one ordinary double point; see [20, Definition 5.1, Remark (b)]. The following product example explains why the stronger local requirement cannot be inserted into Theorem 1.2.

Proposition A.1. *There is a smooth projective fourfold X with pseudo-effective, non-nef canonical divisor for which no ordinary K_X -negative program can reach a nef model while keeping every model analytically locally $\mathbb{Q}$ -factorial. Its unique possible first negative birational contraction is divisorial. Its target is nevertheless globally factorial and admits a nef canonical divisor.*

Proof. Construction. Let $b : B \rightarrow \mathbb{P}^4$ be the blowup of a point, let $H = b^*\mathcal{O}_{\mathbb{P}^4}(1)$, and let F be the exceptional divisor. The graph of projection from that point embeds B in $\mathbb{P}^4 \times \mathbb{P}^3$, and the restriction of $\mathcal{O}(1, 1)$ is $2H - F$. Thus $2H - F$ is very ample. Since H is globally generated,

$$6H - 2F = 2(2H - F) + 2H$$

is very ample as well. Take a general smooth member $U \in |6H - 2F|$. Its image $U_0 \subset \mathbb{P}^4$ is smooth away from the blown-up point. The quadratic jets of the defining sextics at that point are arbitrary,

so a general member has a nondegenerate quadratic initial term. The holomorphic Morse lemma identifies its unique singularity with

$$xy - zw = 0.$$

The restriction $\pi : U \rightarrow U_0$ resolves this node, with exceptional quadric

$$E = F|_U \simeq \mathbb{P}^1 \times \mathbb{P}^1, \quad \mathcal{O}_U(E)|_E = \mathcal{O}_E(-1, -1).$$

Adjunction gives

$$K_U = H|_U + E, \tag{A.1}$$

since $K_B = -5H + 3F$.

Integral weak Lefschetz for the smooth ample threefold $U \subset B$ [19, Section 7, Corollary 7.3] gives

$$H^1(U, \mathbb{Z}) = 0, \quad H^2(U, \mathbb{Z}) = \mathbb{Z}[H|_U] \oplus \mathbb{Z}[E].$$

This is the blowup calculation underlying the classical example in [20, Definition 5.1, Remark (b)]. Let C be a smooth elliptic curve and set $X = U \times C$. Its canonical divisor is the pullback of (A.1), so is effective. If ℓ is either ruling in $E \times \{c\}$, then $K_X \cdot \ell = -1$; in particular K_X is not nef.

All negative contractions have the same target. Put $D = E \times C$ and let $h = \{e\} \times C$ be a horizontal curve. A curve not contained in D has nonnegative degree against both the effective divisor D and the nef hyperplane pullback. Thus every K_X -negative curve lies in D . The two rulings of E have the same class in $H_2(U, \mathbb{R})$: both have degrees 0, -1 against the displayed basis of $H^2(U, \mathbb{R})$. Künneth and $H^1(U, \mathbb{R}) = 0$ now show that every integral curve $\gamma \subset D$ has class

$$[\gamma] = a[\ell] + b[h], \quad a, b \geq 0. \tag{A.2}$$

Here a is the sum of the projection degrees to the two factors of $E = \mathbb{P}^1 \times \mathbb{P}^1$, and b is the projection degree to C . Moreover $K_X \cdot \gamma = -a$, so negativity means $a > 0$.

Let $f : X \rightarrow Z$ be any nontrivial projective birational contraction with connected fibres, normal compact Kähler target, and $-K_X$ relatively ample. Choose a Kähler class ω_Z and set $\alpha = f^*\omega_Z$. For a curve, vanishing of its degree against α is equivalent to being contracted by f . In particular $\alpha \cdot \ell \geq 0$ and $\alpha \cdot h > 0$: the horizontal curve cannot be contracted because $K_X \cdot h = 0$. A contracted curve γ has the form (A.2) with $a > 0$, and

$$0 = \alpha \cdot \gamma = a(\alpha \cdot \ell) + b(\alpha \cdot h).$$

It follows that $b = 0$ and $\alpha \cdot \ell = 0$. Thus all rulings in every $E \times \{c\}$ are contracted, and f is not small.

Set $q = \pi \times \text{id}_C : X \rightarrow U_0 \times C$. Its nontrivial fibres are the quadrics $E \times \{c\}$, which are connected by their rulings, so f is constant on every q -fibre. Conversely, every curve in an f -fibre has the form (A.2) with $b = 0$, and is contracted by q . A connected projective fibre is connected by chains of curves, so q is constant on every f -fibre. The two targets are isomorphic: the reduced image of (q, f) in their product projects properly, bijectively and bimeromorphically to each normal target, hence finitely and isomorphically. Therefore q is the only possible first negative birational contraction. It is divisorial, and its relative numerical rank is one because both quadric rulings have the same nonzero numerical class.

Failure of analytic local $\mathbb{Q}$ -factoriality. Near the singular locus of $U_0 \times C$, consider the prime analytic divisor

$$P = \{x = z = 0\} \times (\text{a disk}) \subset \{xy - zw = 0\} \times (\text{a disk}).$$

On the blowup of the singular axis, its strict transform meets each exceptional quadric in a ruling line. If nP were Cartier near an axis point for some integer $n > 0$, its pullback would have the form $n\tilde{P} + kE_{\text{exc}}$. The associated line bundle restricts trivially to the quadric over that point, because it is pulled back from a line bundle at a point. On the other hand its restriction is

$$\mathcal{O}_{\mathbb{P}^1 \times \mathbb{P}^1}(n - k, -k),$$

up to interchanging the factors. Triviality forces $k = 0$ and $n = 0$, a contradiction. The target is not analytically locally $\mathbb{Q}$ -factorial. Since K_X is not nef and there is no small negative contraction, there is no permissible first step under that local convention.

Compatibility with the global convention. The same example has a valid one-step global program. Indeed, $-E$ is π -ample, so $-K_X$ is q -ample, and adjunction on U_0 gives

$$\mathcal{O}_{U_0 \times C}(K_{U_0 \times C}) \simeq \mathrm{pr}_1^* \mathcal{O}_{U_0}(1),$$

which is nef. On the blowup of the node the exceptional divisor has log discrepancy two; its smooth product with C is a log resolution with the same log discrepancy. Hence the target is terminal, in particular klt.

It remains to check that the local divisor P is not a global factoriality obstruction. The displayed integral cohomology of U and Hodge decomposition give $H^1(U, \mathcal{O}_U) = H^2(U, \mathcal{O}_U) = 0$ and $\mathrm{Pic}(U) = \mathbb{Z}[H|_U] \oplus \mathbb{Z}[E]$. The exponential sequence and Künneth therefore give

$$\mathrm{Pic}(X) = \mathbb{Z}[H] \oplus \mathbb{Z}[D] \oplus \mathrm{pr}_C^* \mathrm{Pic}(C).$$

Strict transform and divisor pushforward identify

$$\mathrm{Cl}(U_0 \times C) = \mathrm{Pic}(X)/\mathbb{Z}[D].$$

Every surviving class descends to a Cartier class, either from the hyperplane bundle on U_0 or from a line bundle on C . Chow's theorem makes every global analytic Weil divisor on this projective target algebraic, so every such divisor is Cartier [21, Section 4, no. 19, Proposition 13]. GAGA identifies its global coherent rank-one reflexive sheaves with algebraic divisorial sheaves, which are therefore invertible as well [21, Section 3, no. 12, Theorems 2–3]. The plane P is defined only on an analytic neighbourhood; it need not extend to a divisor on the whole compact space. $\square$

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